Máster de Formación Permanente

POLITÉCNICA | UNIVERSIDAD POLITÉCNICA DE MADRID | escuela técnica superior de ingeniería diseño industrial | POLITÉCNICA Máster ERMA

Máster en Energías Renovables y Medio Ambiente

20 EDICIÓN 2025-2026



Módulo 15
Plantas híbridas y de almacenamiento

Centrales reversibles: Análisis de Proyectos existentes

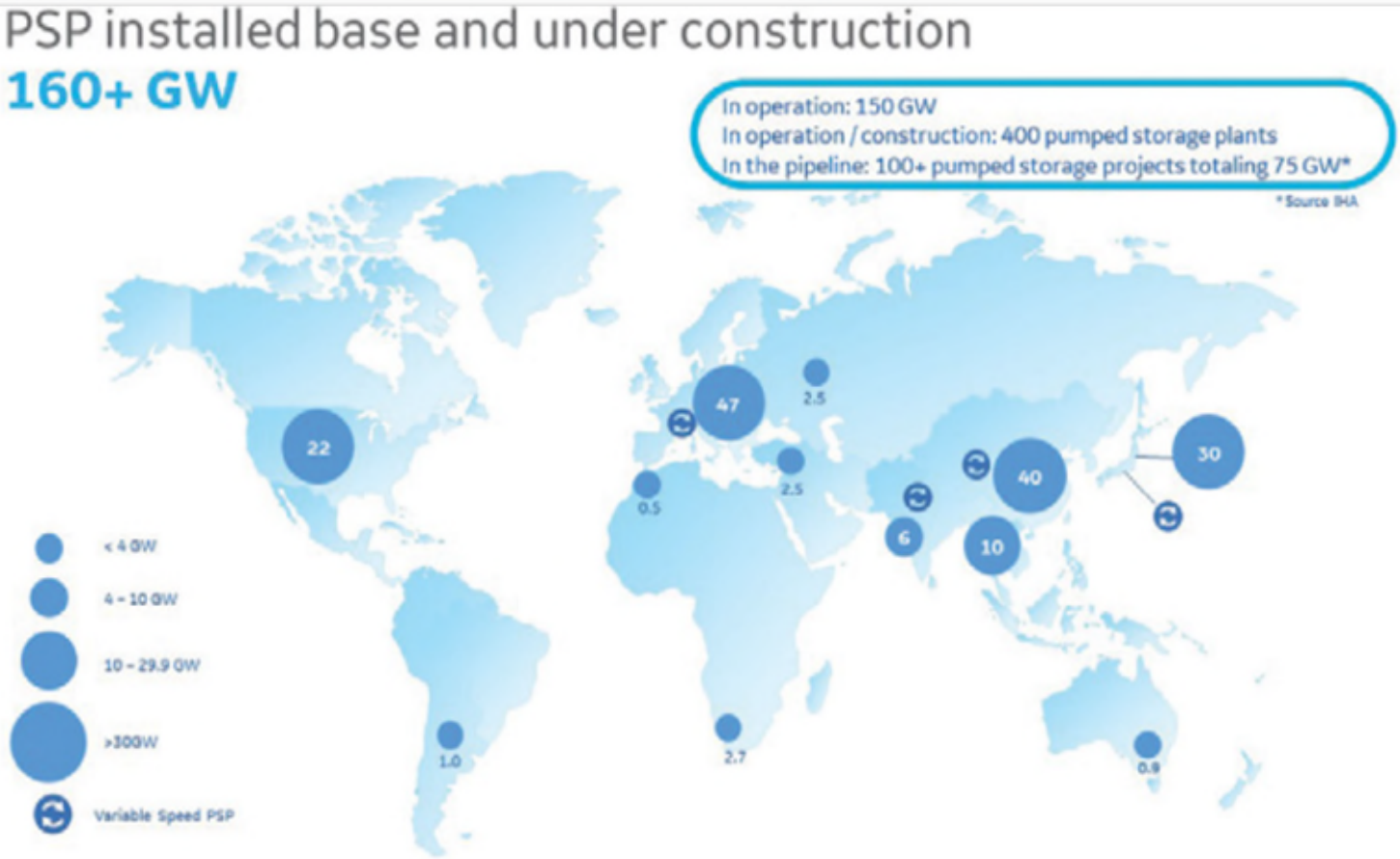
Jose Manuel Clamagirand (SENER)

INDICE GENERAL

1. **INTRODUCCION**
2. PROYECTOS NIVEL NACIONAL
3. PROYECTOS NIVEL INTERNACIONAL

1.-INTRODUCCION

1.-GENERAL



1.-INTRODUCCION

1.- GENERAL

-POWER PLANTS ANDRITZ (MANUFACTURER)

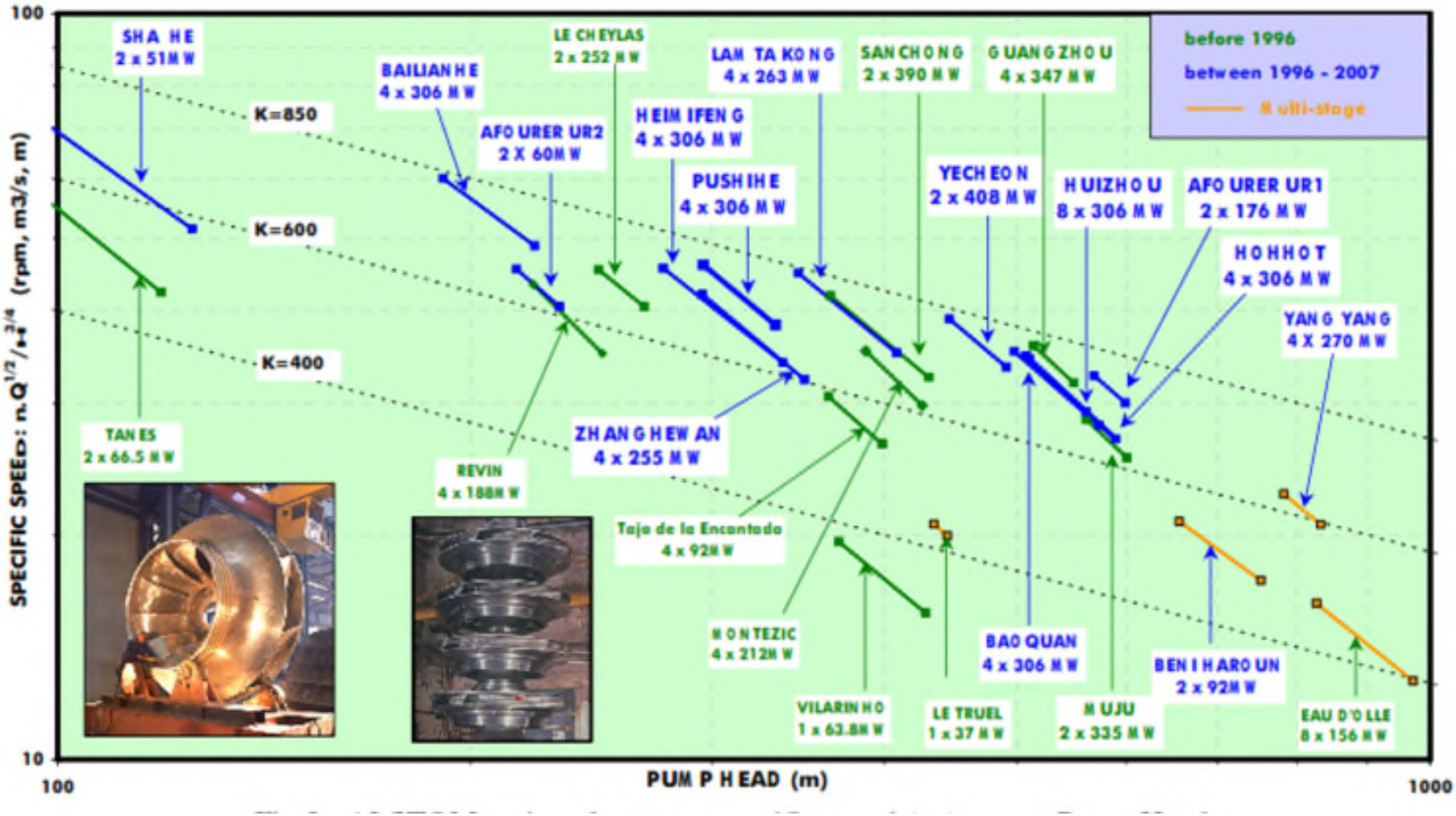
Successful designs of Pump turbines

Projekt	Head [m]	Power [MW]	Runner diameter [mm]	Speed [rpm]	Country	Status
Zarnowiec	127.5	187.6	6008	166.67	Poland	In Operation
Tongbai	288.8	306.0	4802	300	China	In Operation
Lang Ya Shan	152.9	165.7	4700	230.77	China	In Operation
Nestil	1065.7	141.2	2263	600	Switzerland	In Operation
Yixing	420	262	4394	375	China	In Operation
Hintermuhr	517.0	71.9	1870	1000	Austria	In Operation
Feldsee	547.8	72.7	1919	1000	Austria	In Operation
Baixo Sabor Montante	104.6	76.9	4112	214.29	Portugal	Hydraulic Design
Baixo Sabor Jusante	35.2	17.9	3948	150	Portugal	Hydraulic Design
Vianden M11	294.6	200.4	4286	333.33	Luxembourg	Hydraulic Design

1.-INTRODUCCION

1.- GENERAL

-POWER PLANTS ALSTOM (MANUFACTURER)



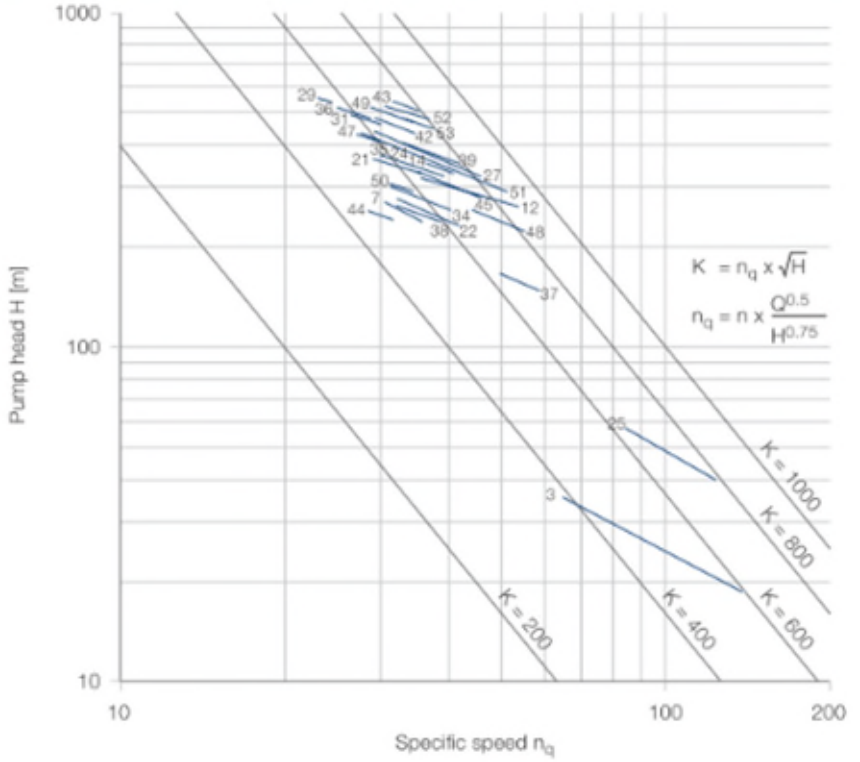
1.-INTRODUCCION

1.- GENERAL

-POWER PLANTS VOITH (MANUFACTURER)

Selected pump-turbines: operation range in pump mode

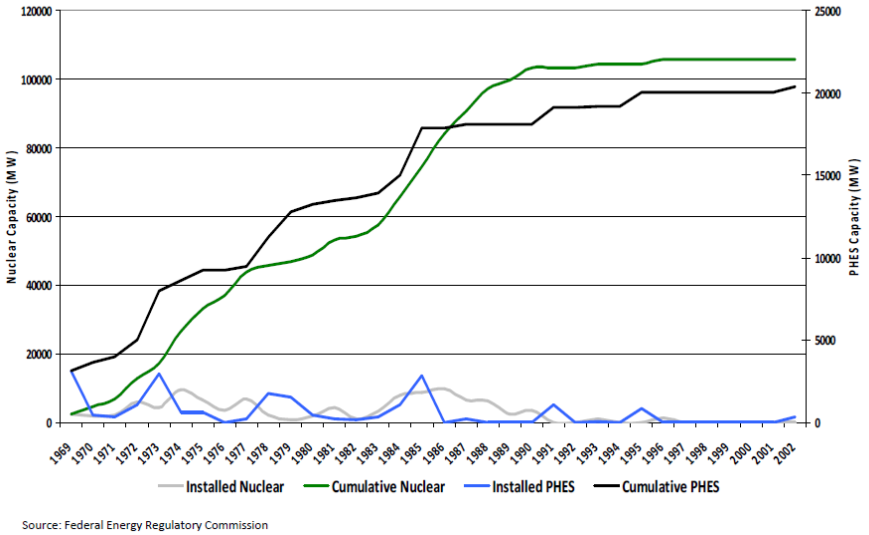
3 Lewiston
7 Coo I
12 Raccoon
14 Rodund II
21 Bath County
22 Coo II
24 Estangento
25 Gabriel y Galan
27 Kühtai
29 Obrovac
31 Presenzano
34 Palmiet
35 Bad Creek
36 La Muela
37 Herdecke
38 Chiotas (4 stage)
39 Mingtan
42 Shisanling
43 Guangzhou II
44 Edolo (5 stage)
45 Goldisthal
(two variable speed)
47 Venda Nova
48 Tai An
49 Siah Bishe
50 Waldeck I
51 Limberg II
52 La Muela II
53 Ingula



1.-INTRODUCCION

2.- USA

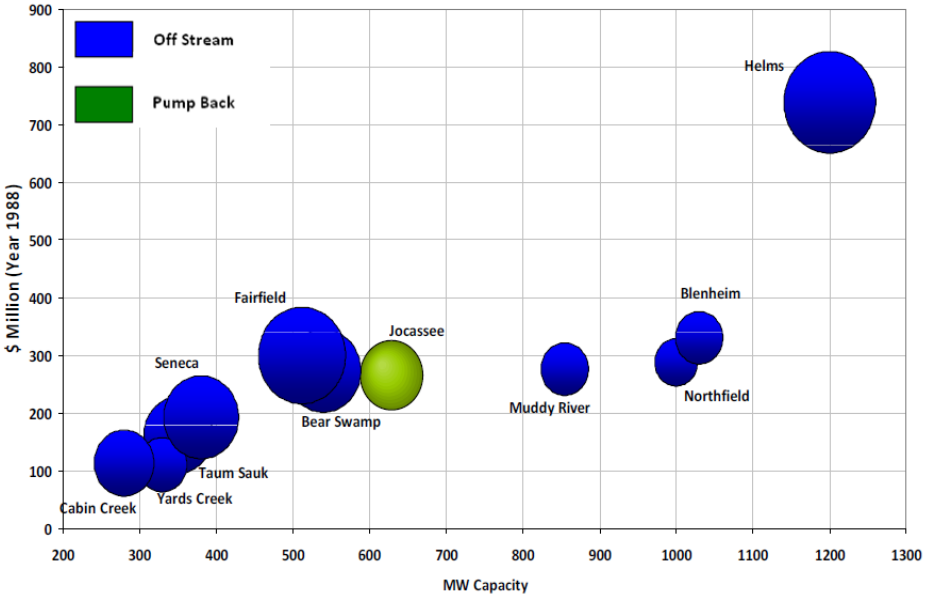
Development of PHES in the USA



Source: Federal Energy Regulatory Commission

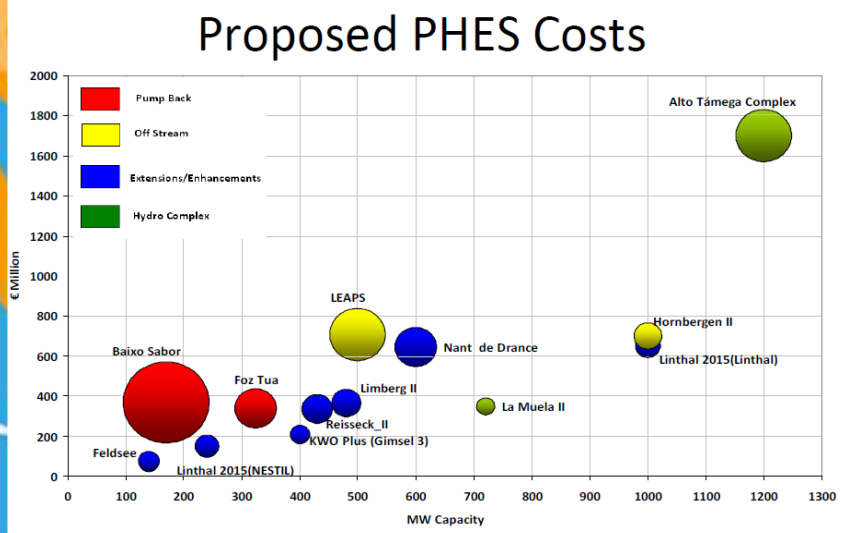
INSTALLED NUCLEAR
~
INSTALLED PHES

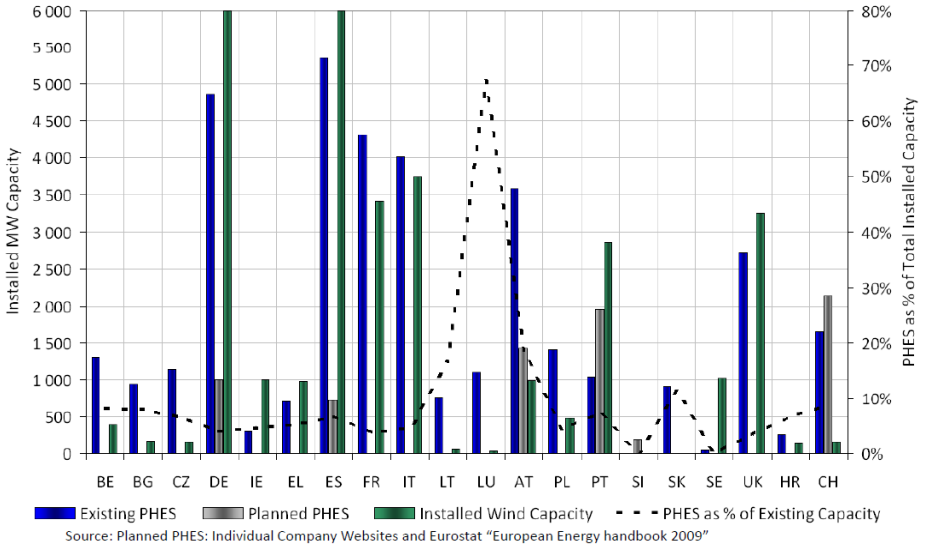
RANGE 288-639 $/Kw (1988)

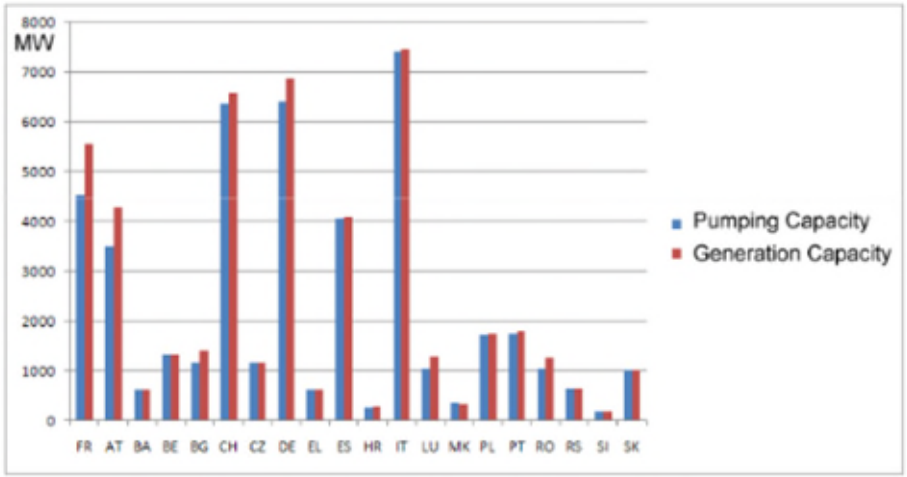


1.-INTRODUCCION

3.- UE







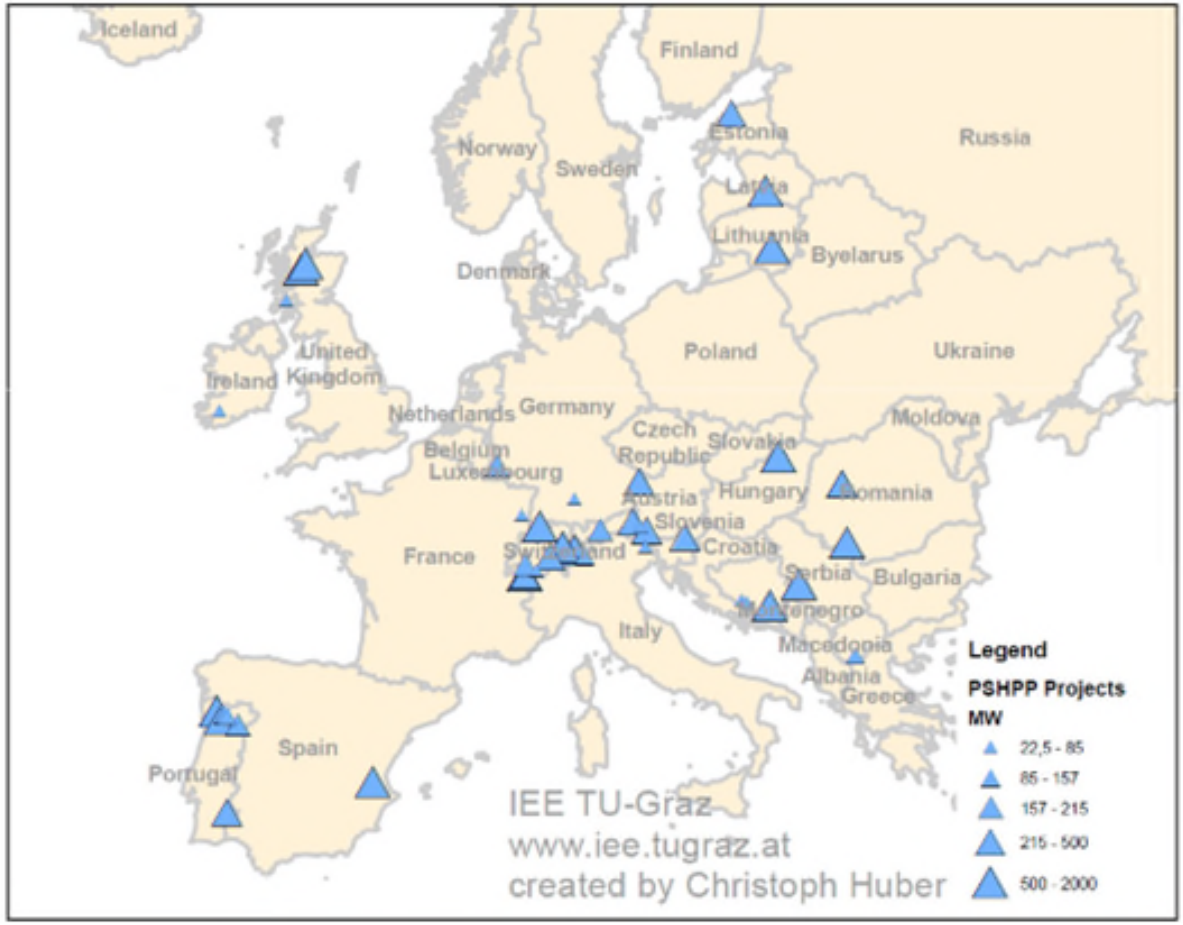
- Wide Range of Costs 470- 2170 €/kW. Very Site/Project Specific
- Over 7 GW of new PHES proposed in Europe

1. Amler, P. Large Hydro ensuring Grid stability with rapidly expanding Wind Generation- A supplier Perspective. Andritz Hydro. May 2010. All Energy 2010 Aberdeen

2. Deane J.P Techno Economic Review of Existing and New Pumped Hydro Energy Storage Plant

1.-INTRODUCCION

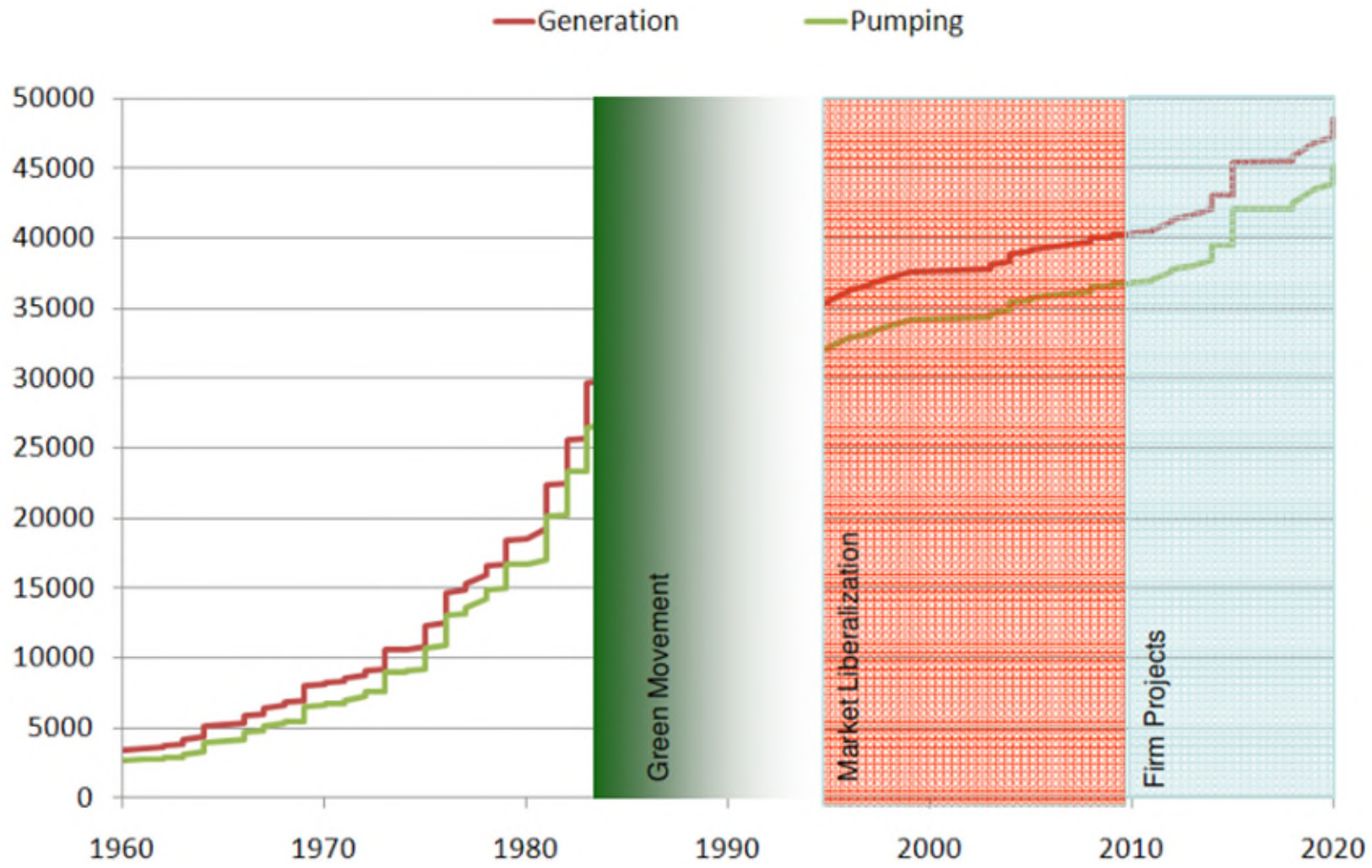
2.- UE
PLANNED CONSTRUCTION



1. Amler, P. Large Hydro ensuring Grid stability with rapidly expanding Wind Generation- A supplier Perspective. Andritz Hydro. May 2010. All Energy 2010 Aberdeen

1.-INTRODUCCION

2.- UE



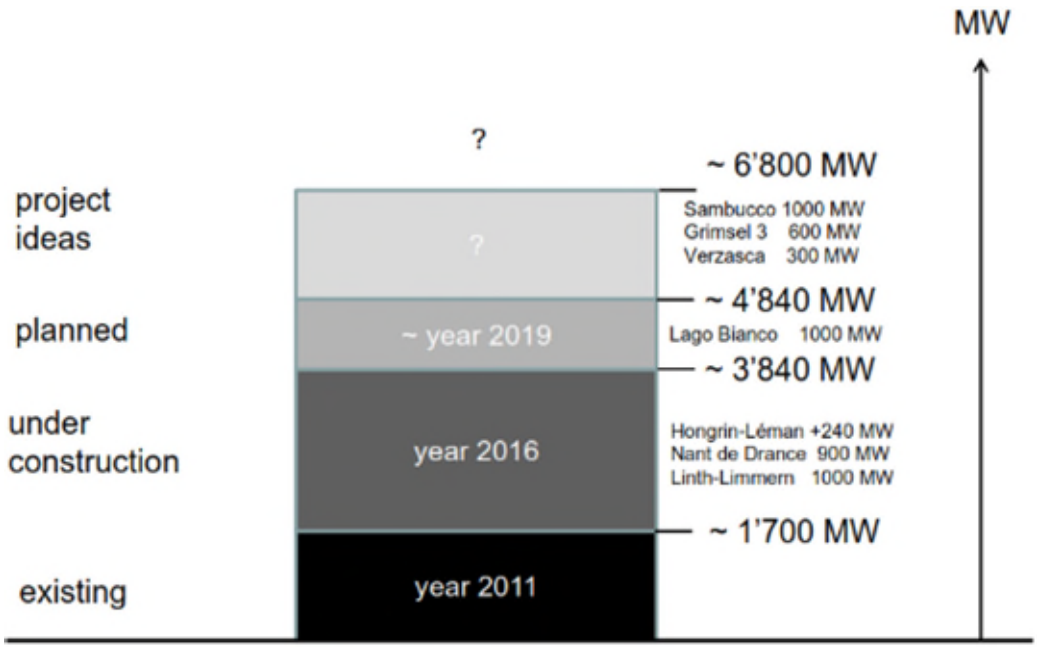
1. Amler, P. Large Hydro ensuring Grid stability with rapidly expanding Wind Generation- A supplier Perspective. Andritz Hydro. May 2010. All Energy 2010 Aberdeen

6.-PHES AMBITO INTERNACIONAL

2.- UE

COUNTRY	TURBINES	PUMPS
GERMANY	6.675 MW	6.220 MW
AUSTRIA	3.887MW	N/A
SWITZERLAND	1.496 MW	990 MW

i.e. : SWITZERLAND



3. Roth, M. Renewable Energy Integration: Its impact on pumped storage power plants. 9/11/12. EWZ
4. Felix, D. Hydropotencial, Pumped Storage and Hydro-Specific Issues. Seminar Frontiers in energy research. May 2012

INDICE GENERAL

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2. **PROYECTOS NIVEL NACIONAL**
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2.-PROYECTOS AMBITO NACIONAL

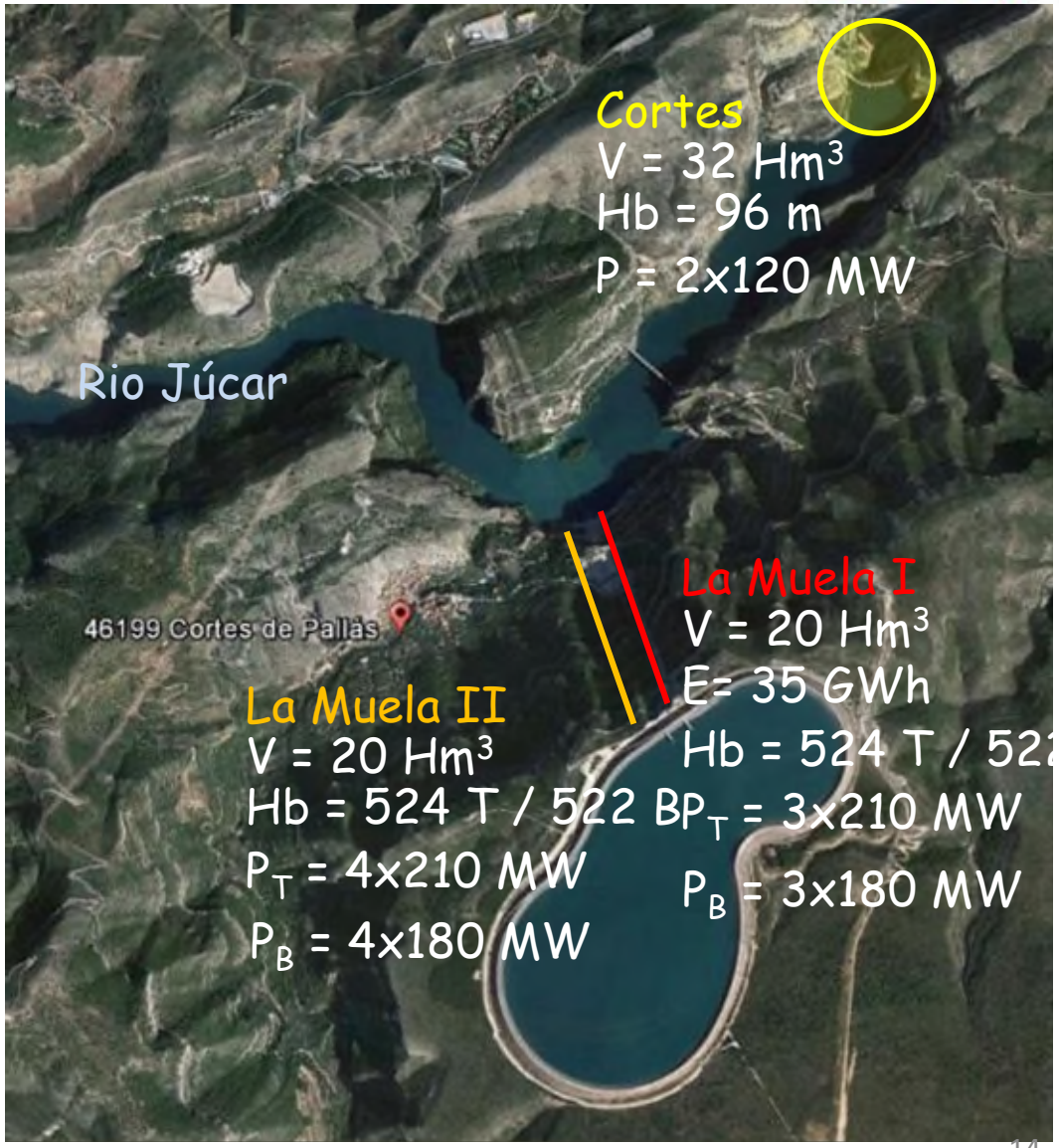


7.-PHES AMBITO NACIONAL

CORTES - LA MUELA I y II (IBERDROLA)

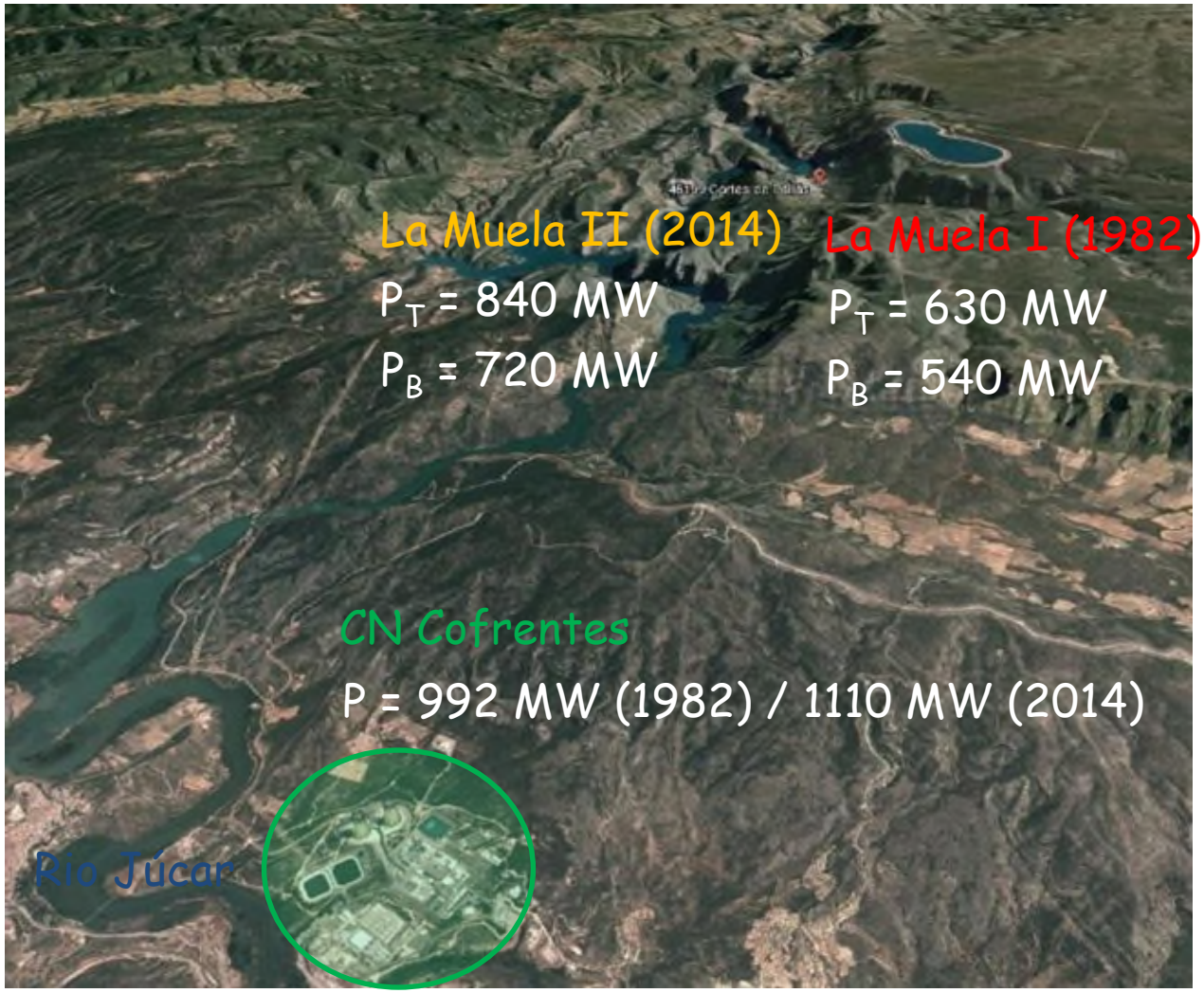






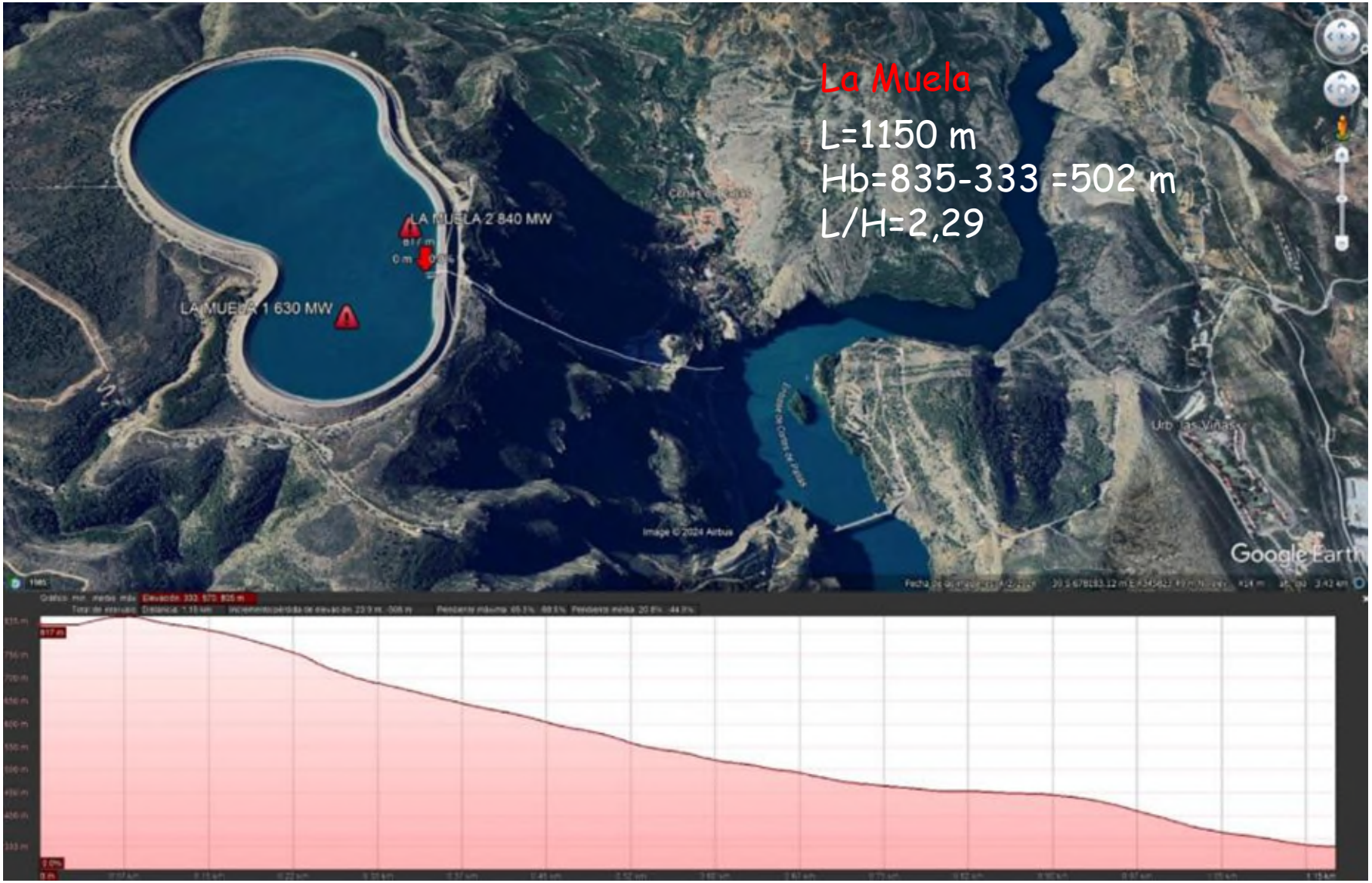
7.-PHES AMBITO NACIONAL

LA MUELA I y II - COFRENTES



7.-PHES AMBITO NACIONAL

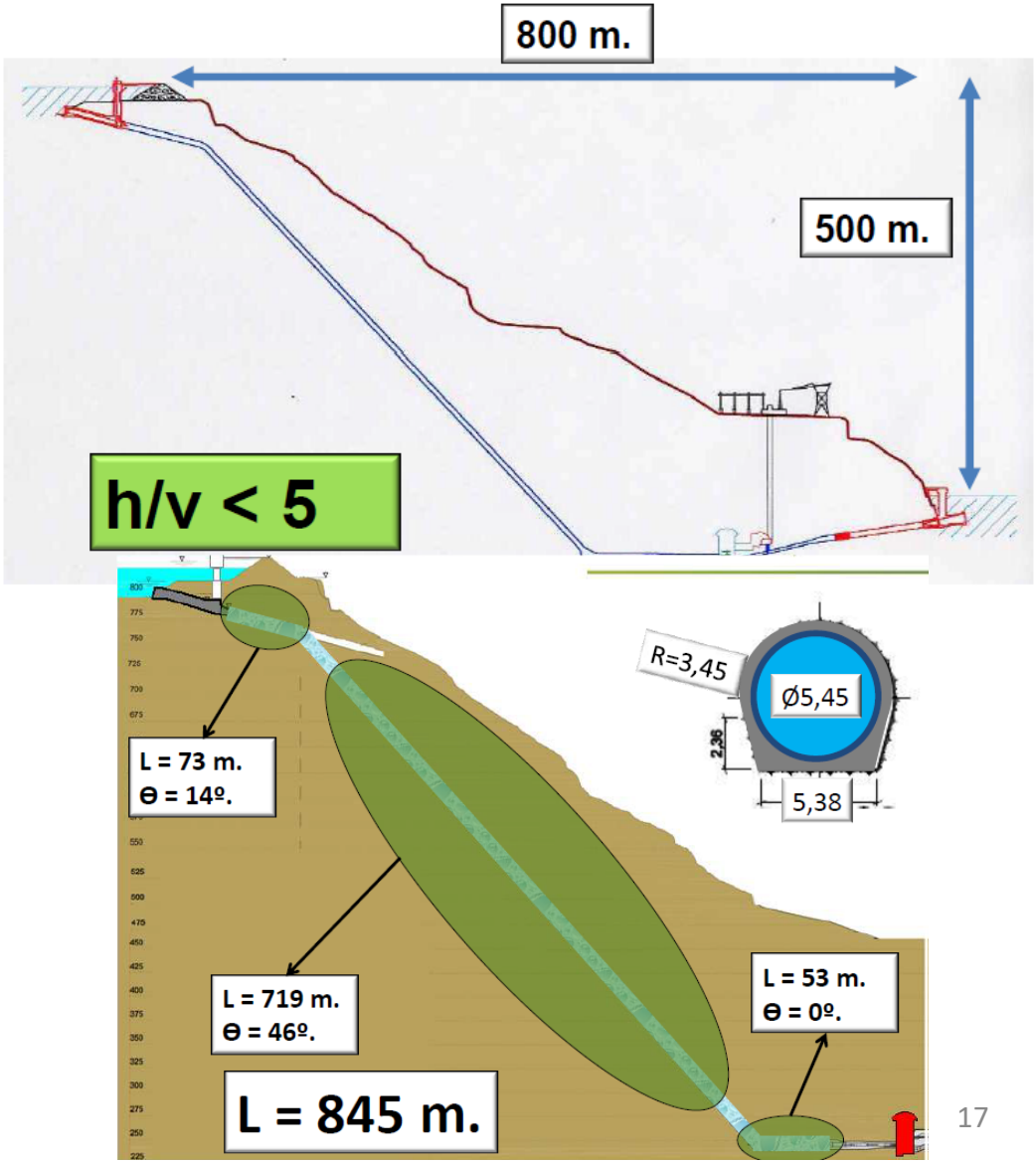
LA MUELA I y II - COFRENTES



7.-PHES AMBITO NACIONAL

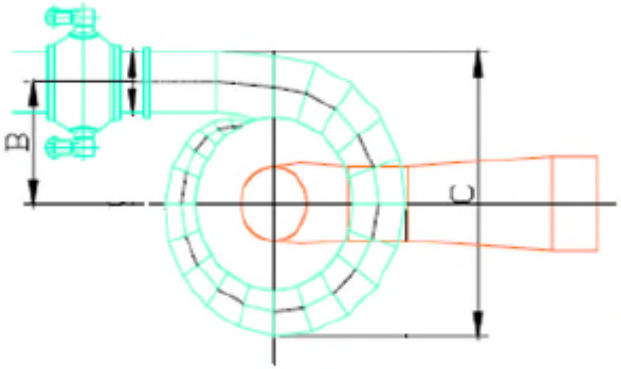
LA MUELA I y II - COFRENTES





7.-PHES AMBITO NACIONAL

LA MUELA I y II - COFRENTES



	LA MUELA I (500 rpm)	LA MUELA II (600 rpm)
B	3.885	3.360 (-24%)
C	9.000	7.981 (-22%)



7.-PHES AMBITO NACIONAL

AGUAYO (VIESGO)

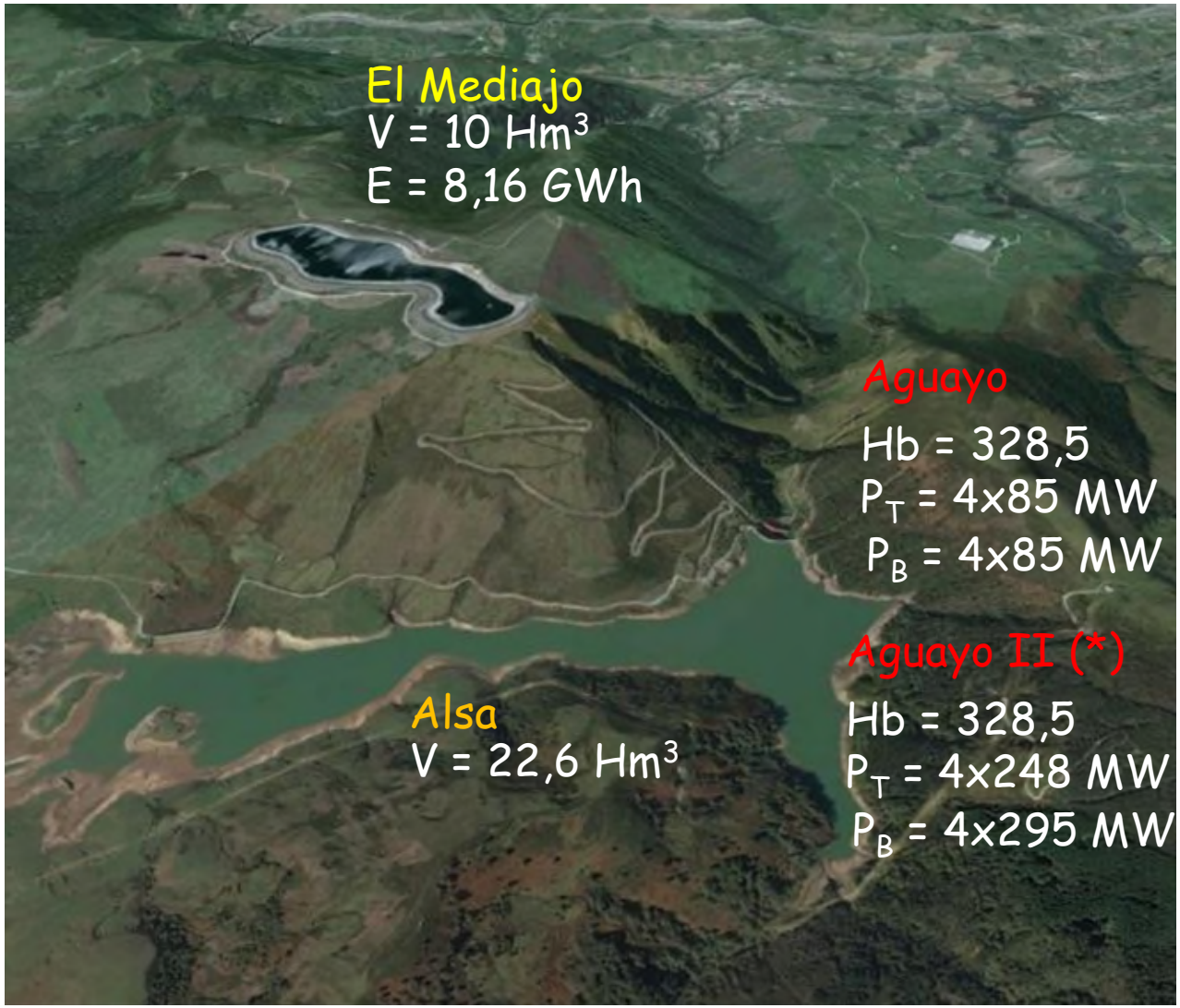






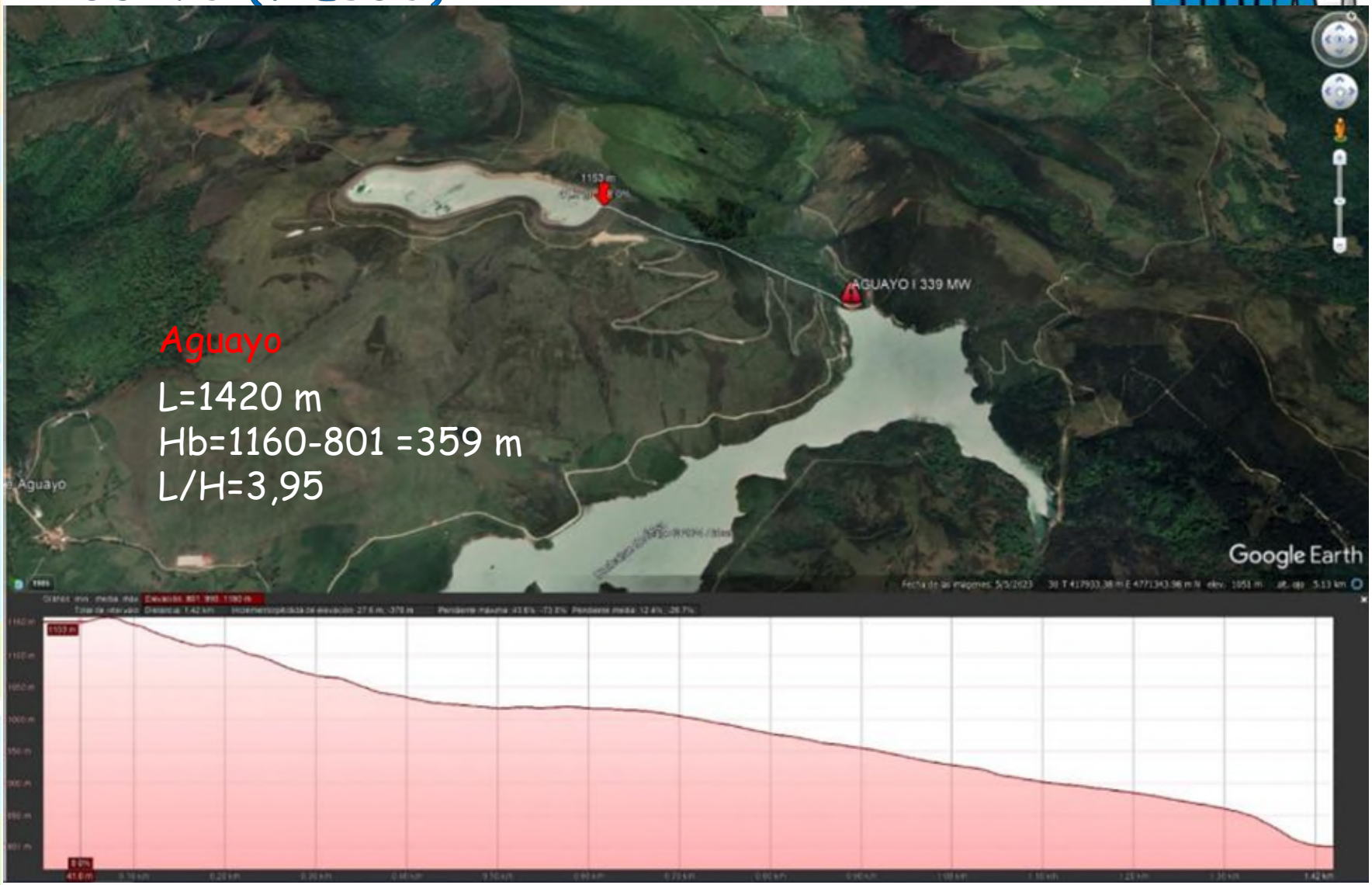
7.-PHES AMBITO NACIONAL

AGUAYO (VIESGO)



7.-PHES AMBITO NACIONAL

AGUAYO (VIESGO)

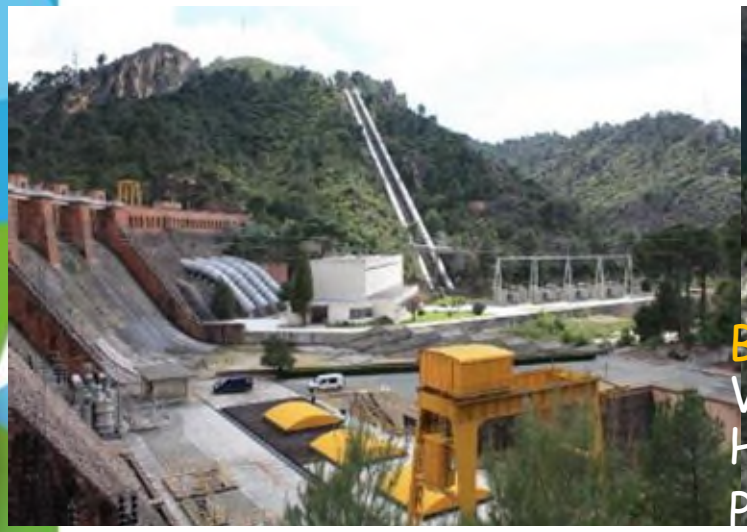


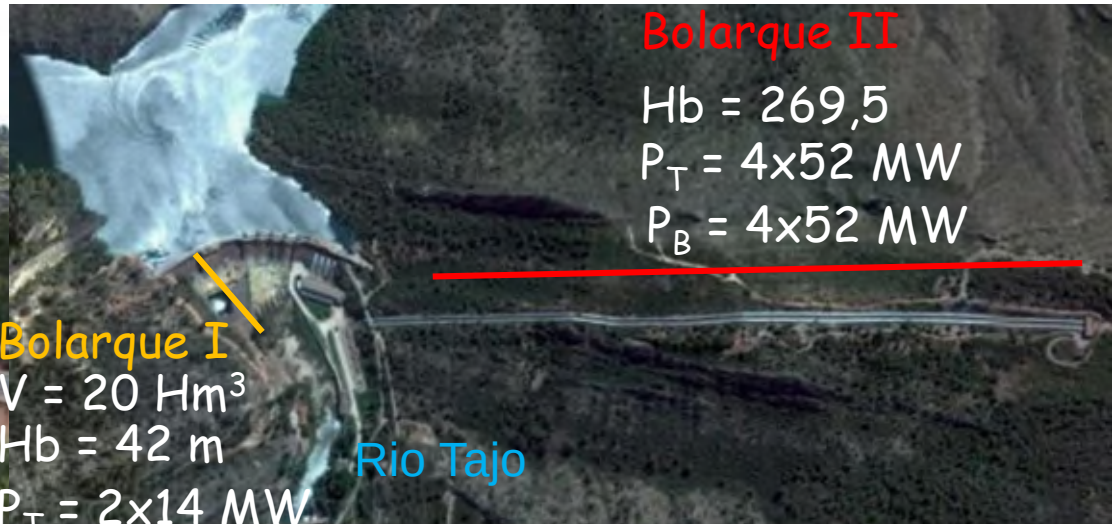
7.-PHES AMBITO NACIONAL

BOLARQUE II (GAS NATURAL UNION FENOSA)



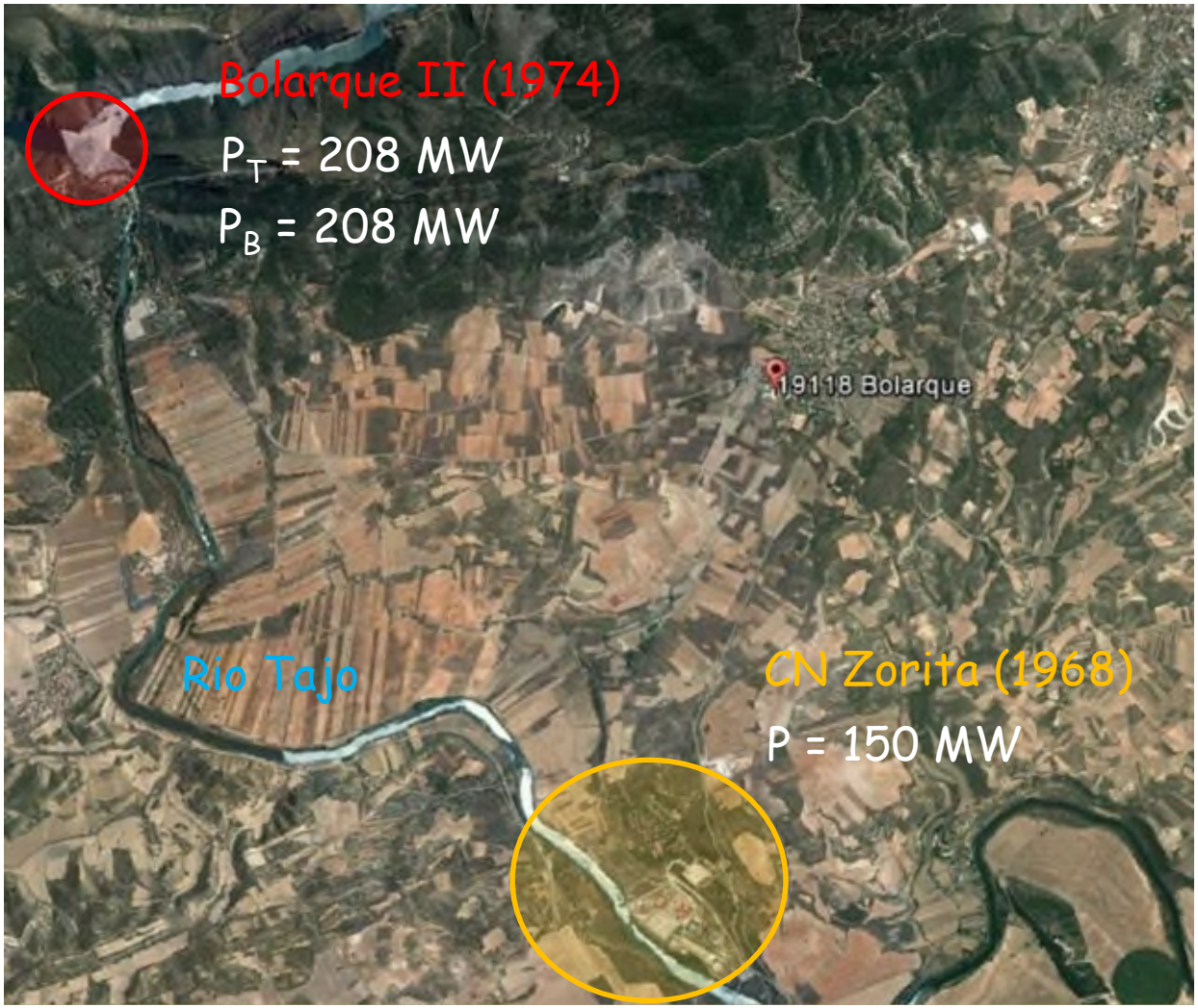






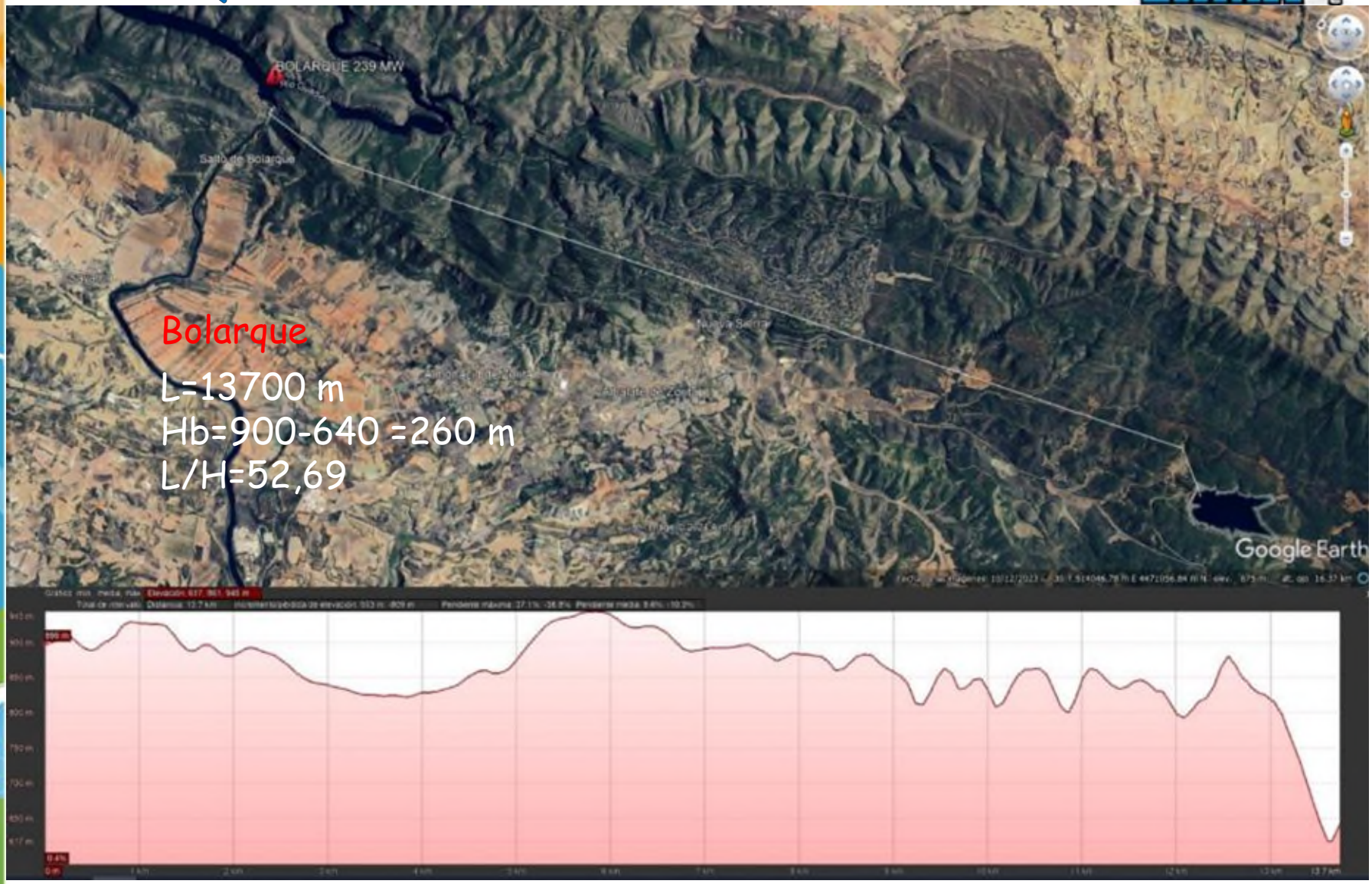
7.-PHES AMBITO NACIONAL

BOLARQUE II - ZORITA



7.-PHES AMBITO NACIONAL

BOLARQUE II - ZORITA



7.-PHES AMBITO NACIONAL

VILLARINO – ALDEAVILA (IBERDROLA)

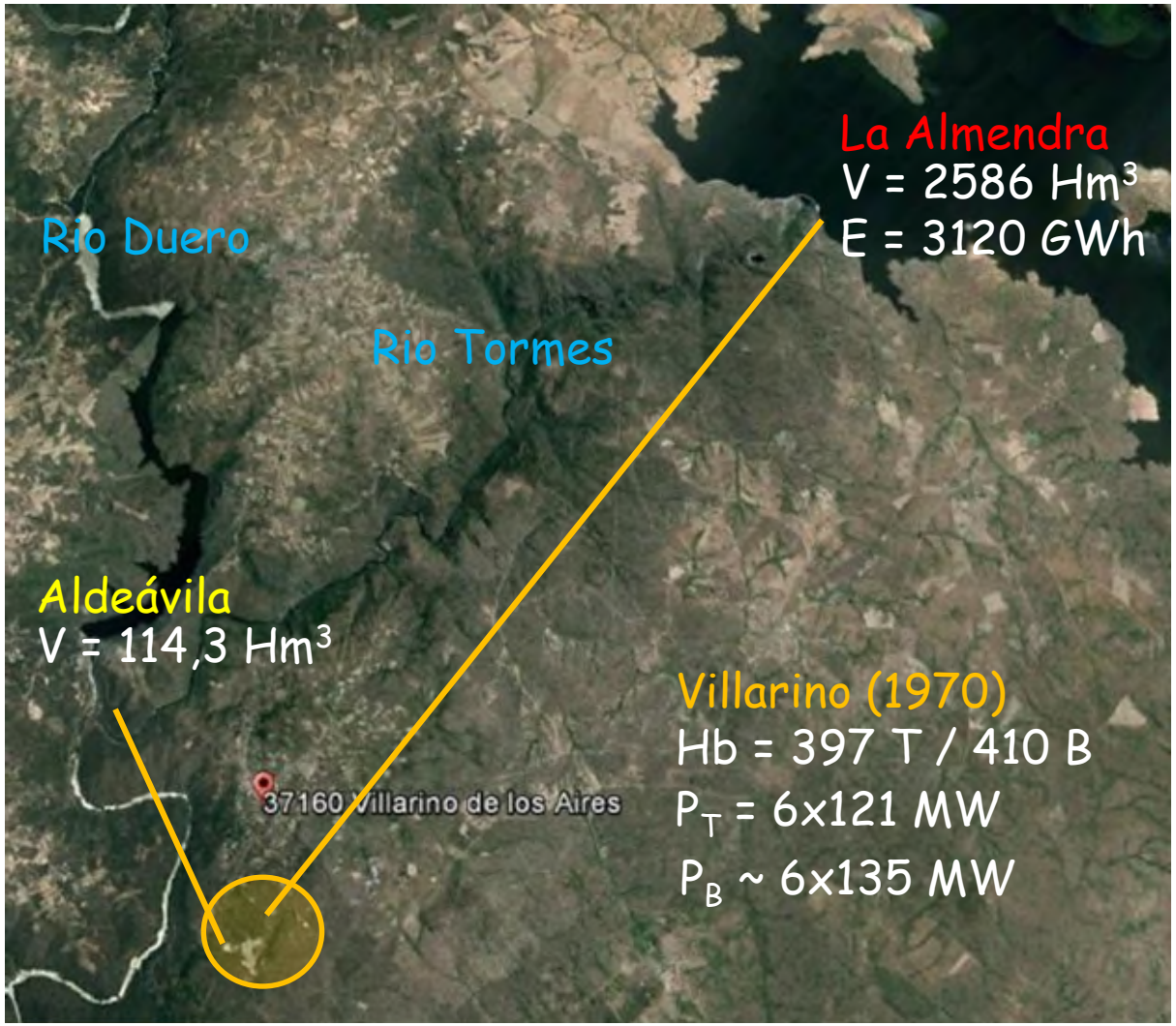






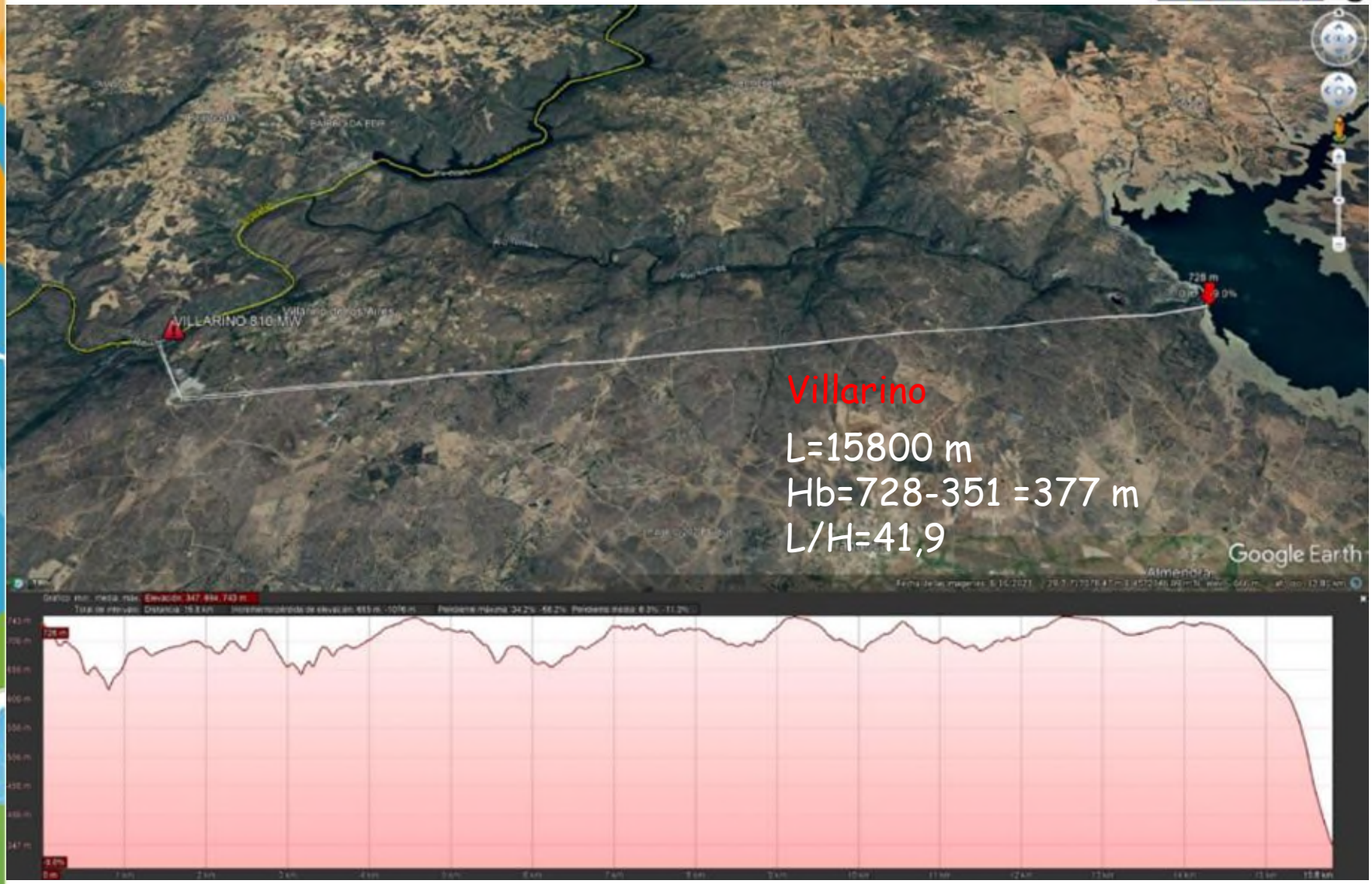
7.-PHES AMBITO NACIONAL

VILLARINO - ALDEÁVILA



7.-PHES AMBITO NACIONAL

VILLARINO - ALDEÁVILA



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3.-PROYECTOS AMBITO INTERNACIONAL

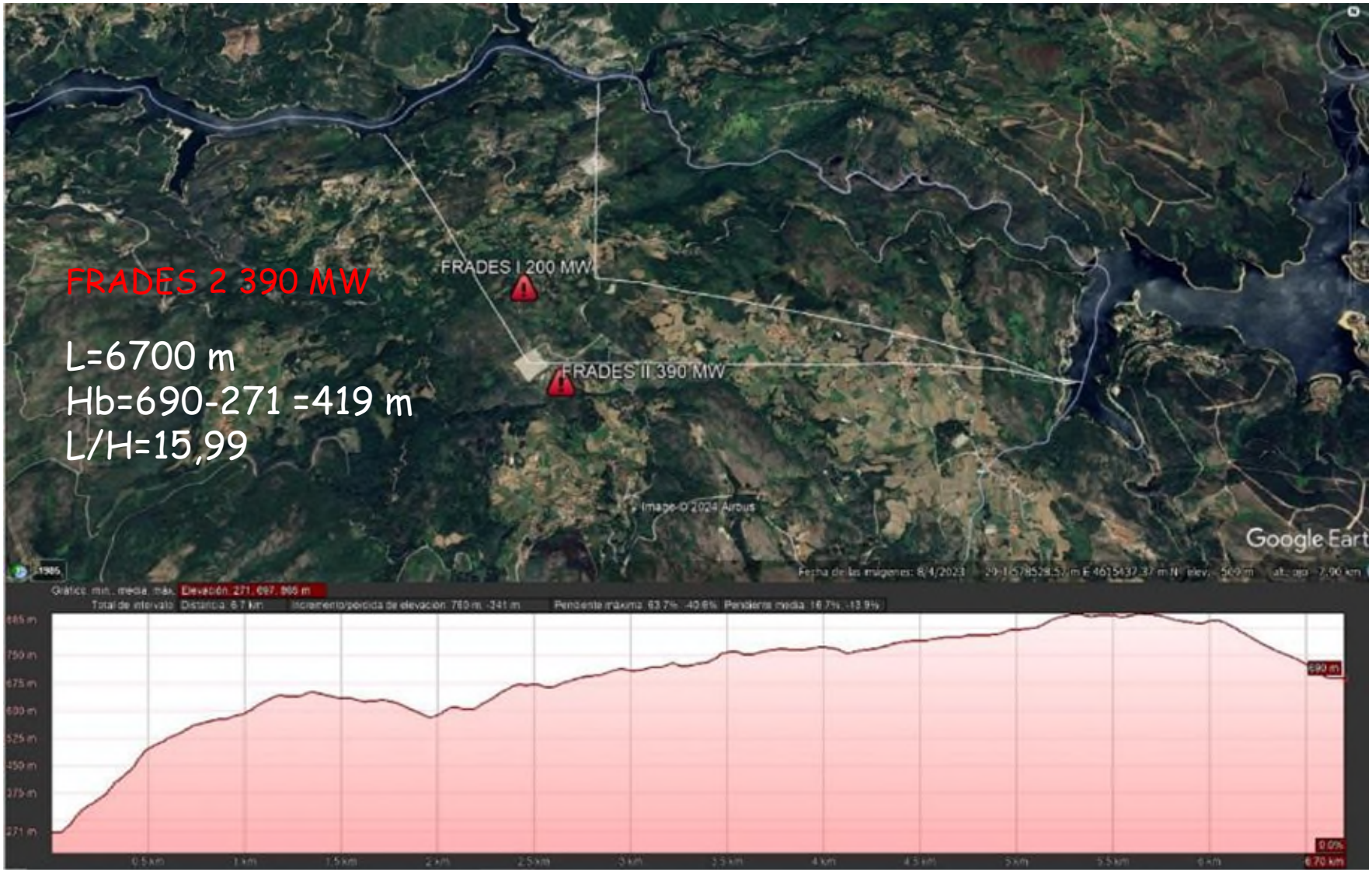
PORTUGAL



3.-PROYECTOS AMBITO INTERNACIONAL

PORTUGAL

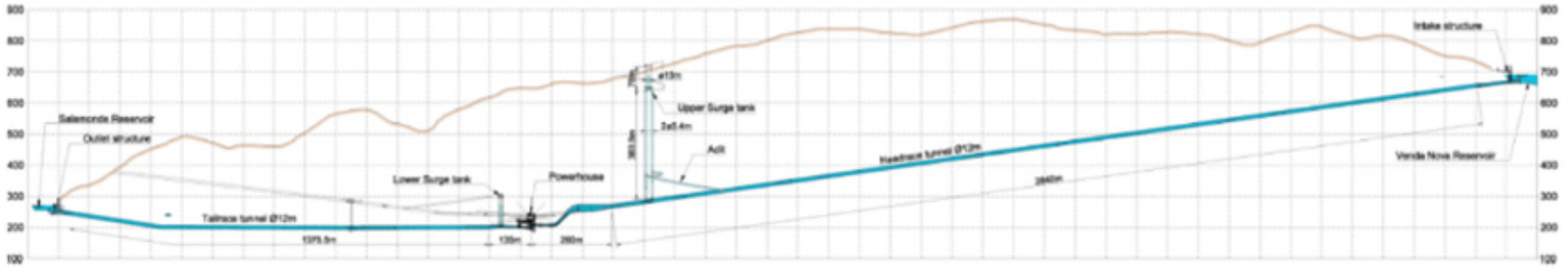
FRADES II

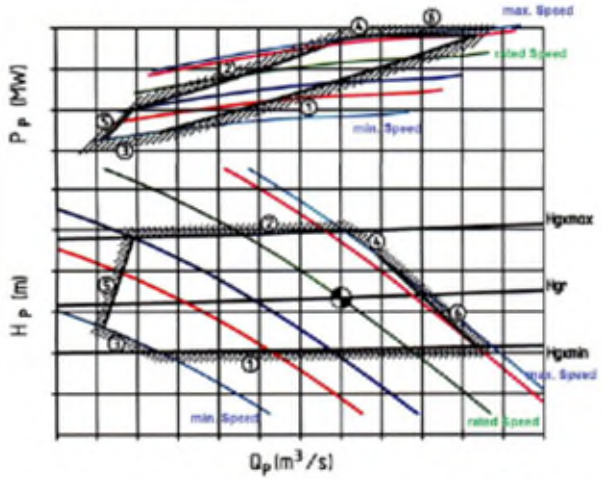


3.-PROYECTOS AMBITO INTERNACIONAL

PORTUGAL

FRADES II





In pump mode the operation range is defined by:

(1) minimum gross head

(2) maximum gross head

(3) minimum speed

(4) maximum speed

(5) pump instability

(6) maximum pump input

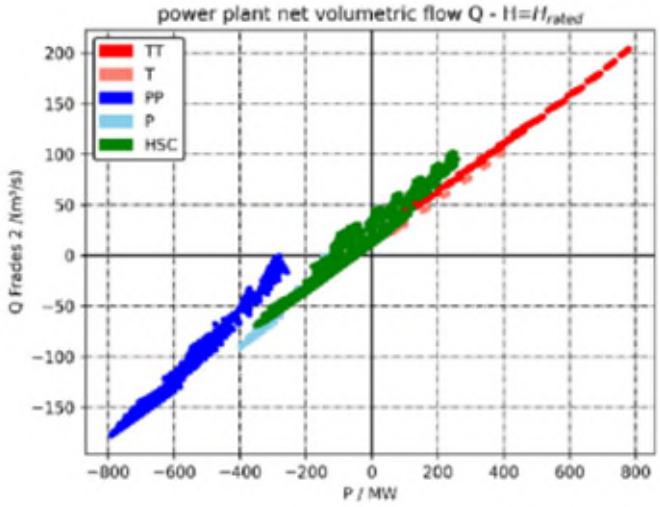
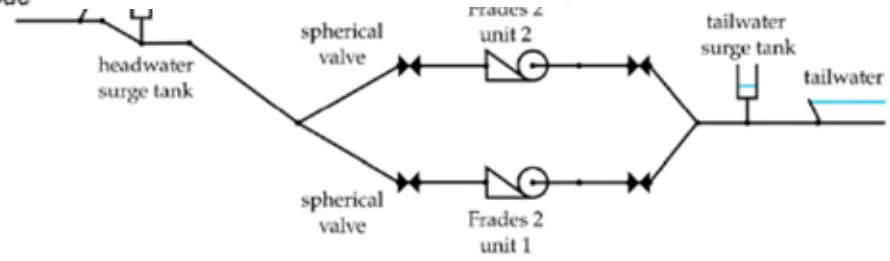


Figure 12: Characteristics for adjustable speed in pump mode



5. 2023 Geiger et all. **Power Plant Transients including Hydraulic Short Circuit Operation Mode**

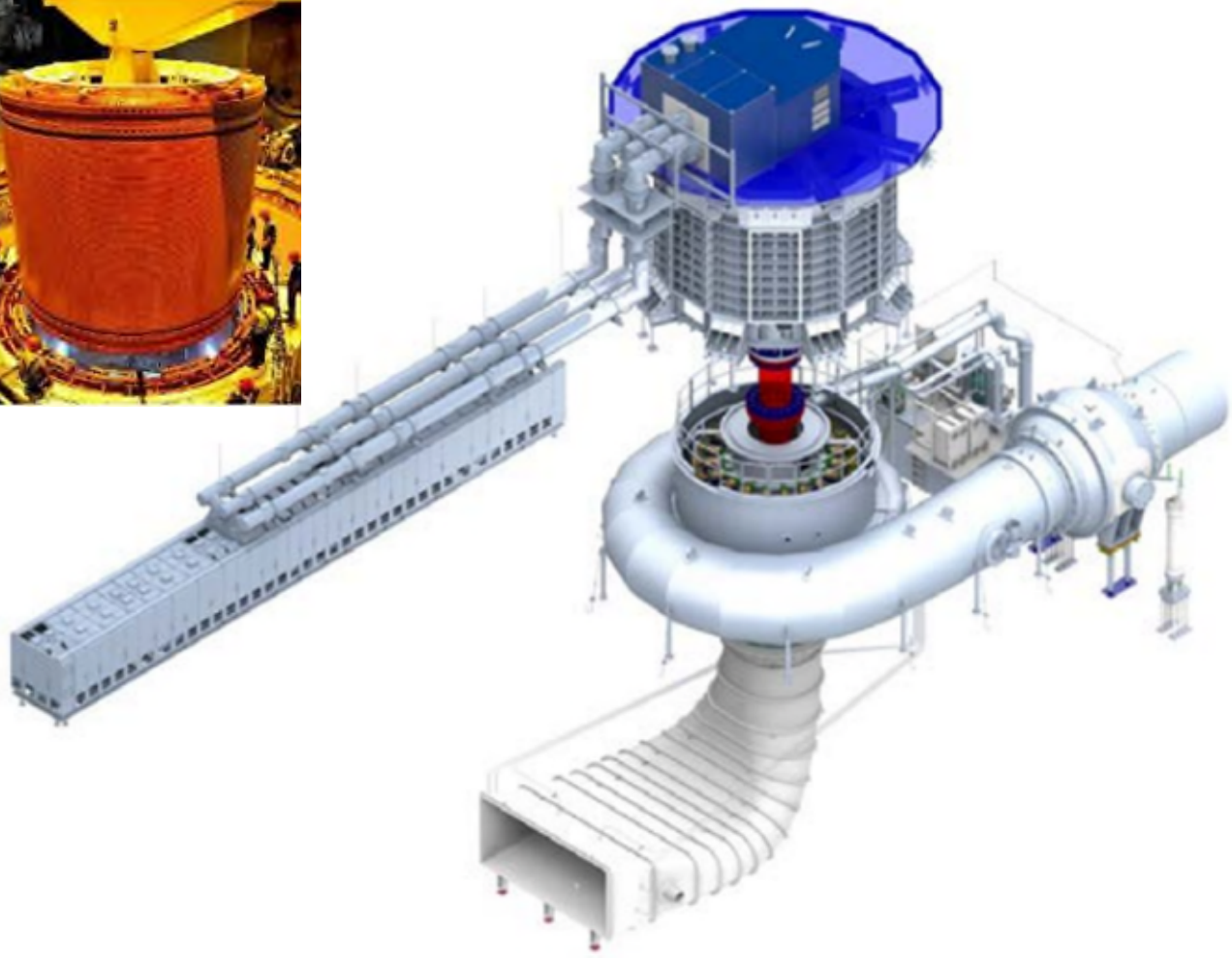
6. 2019 Voith. Recent developments

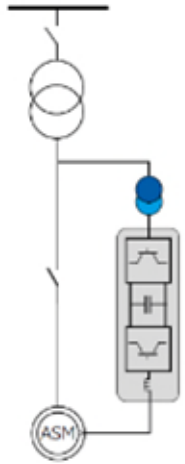
3.-PROYECTOS AMBITO INTERNACIONAL

PORTUGAL

FRADES II



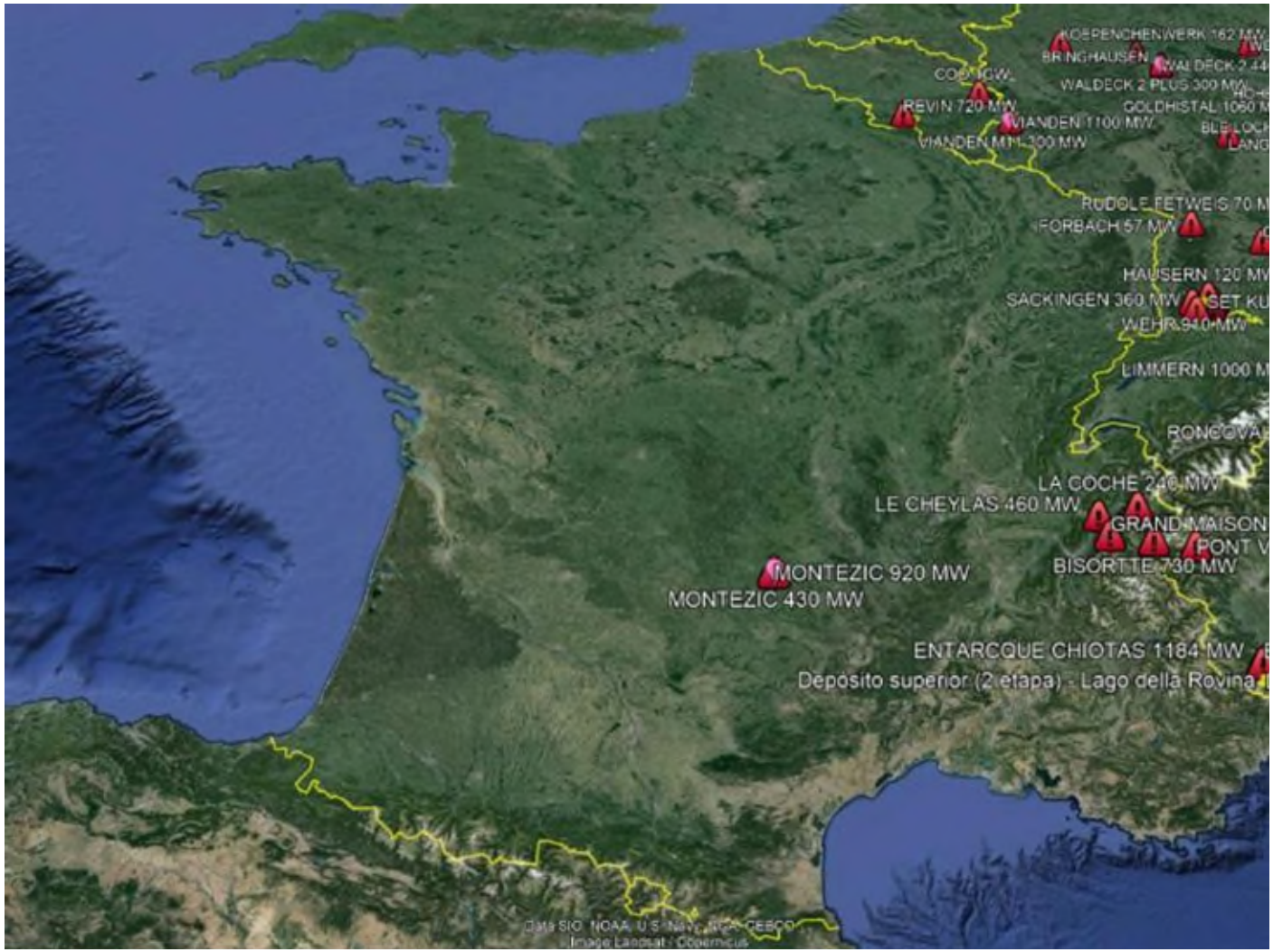




5. 2023 Geiger et all. **Power Plant Transients including Hydraulic Short Circuit Operation Mode**
6. 2019 Voith. Recent developments

3.-PROYECTOS AMBITO INTERNACIONAL

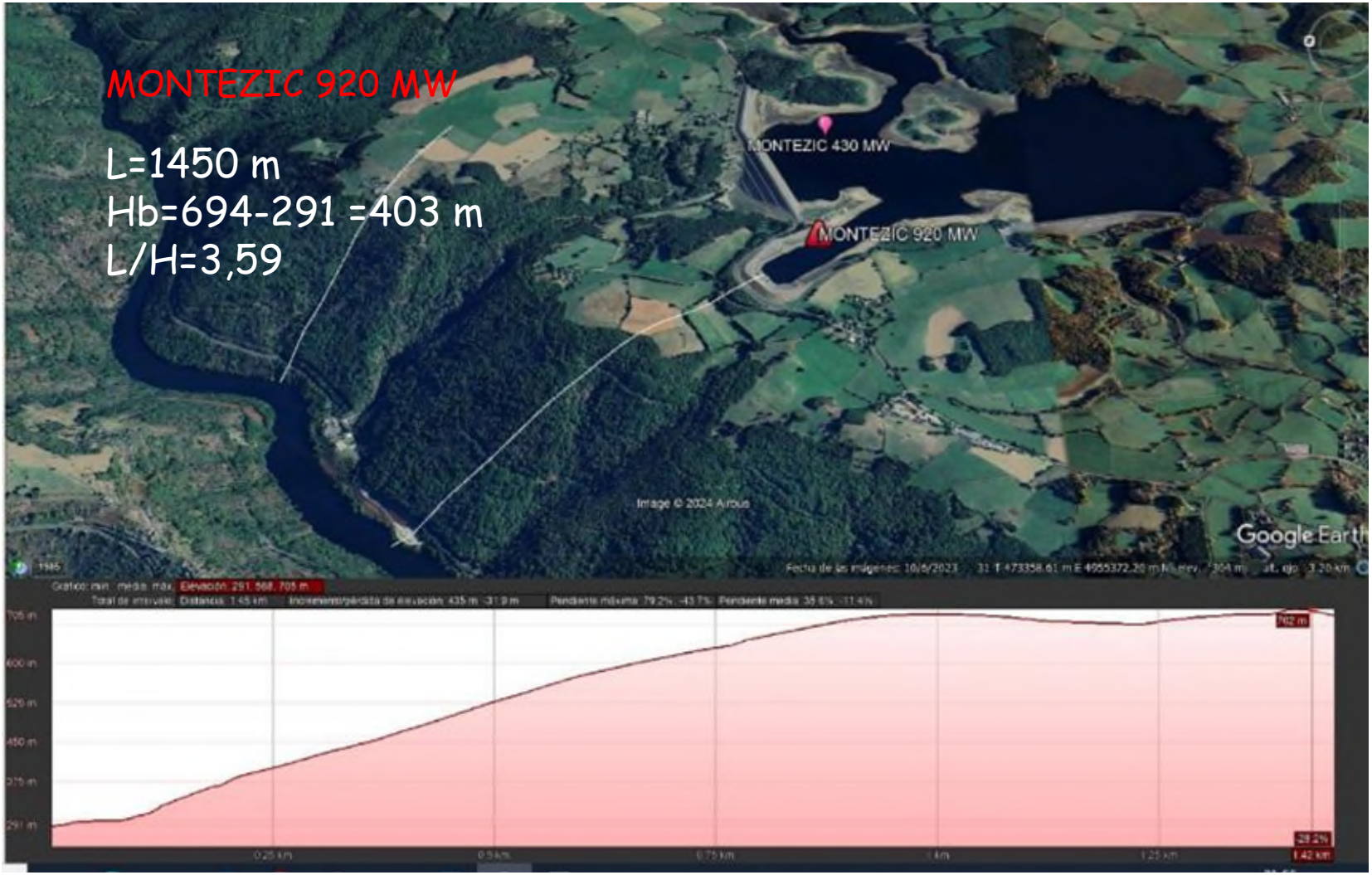
FRANCIA



3.-PROYECTOS AMBITO INTERNACIONAL

FRANCIA

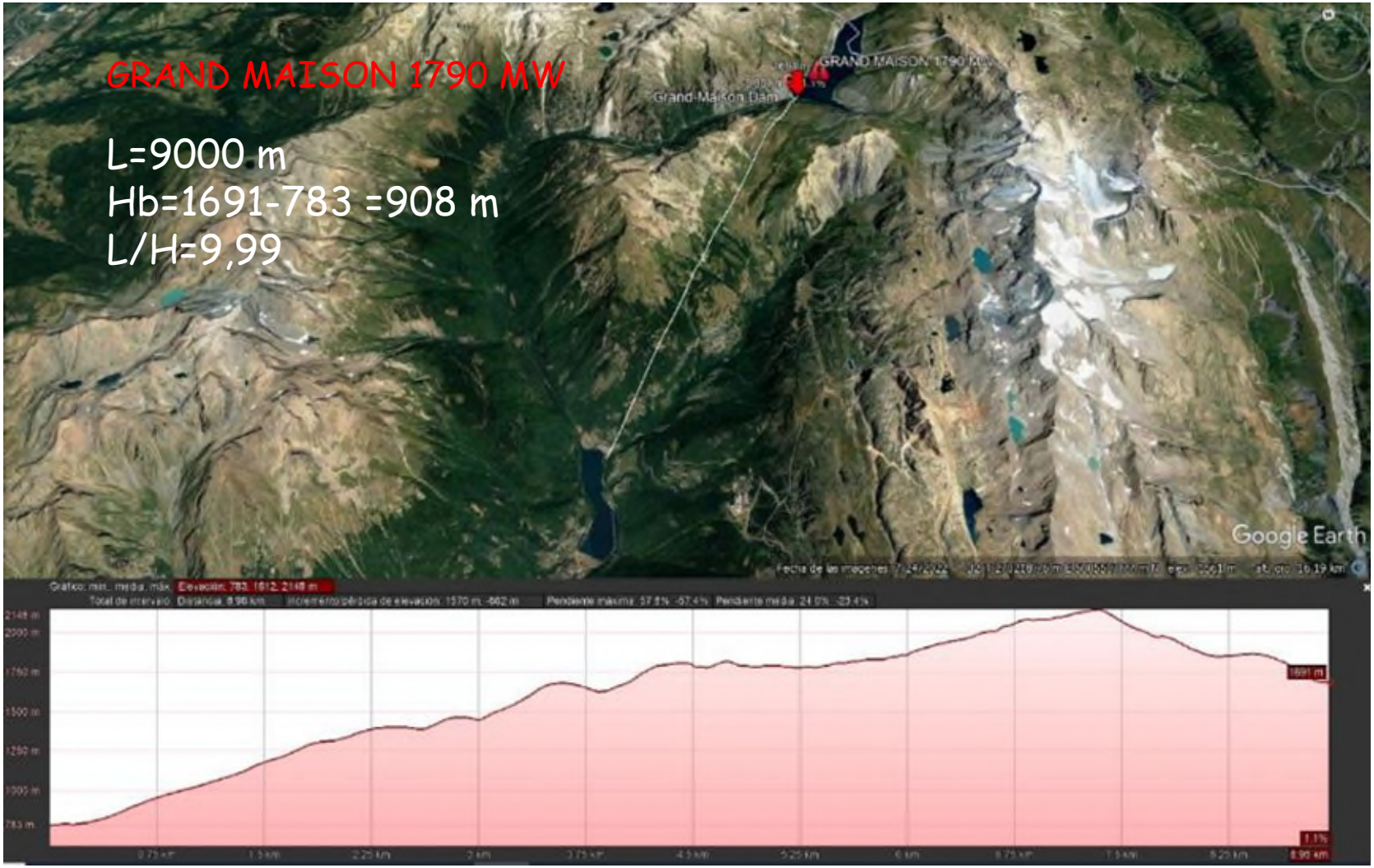
MONTEZIC



3.-PROYECTOS AMBITO INTERNACIONAL

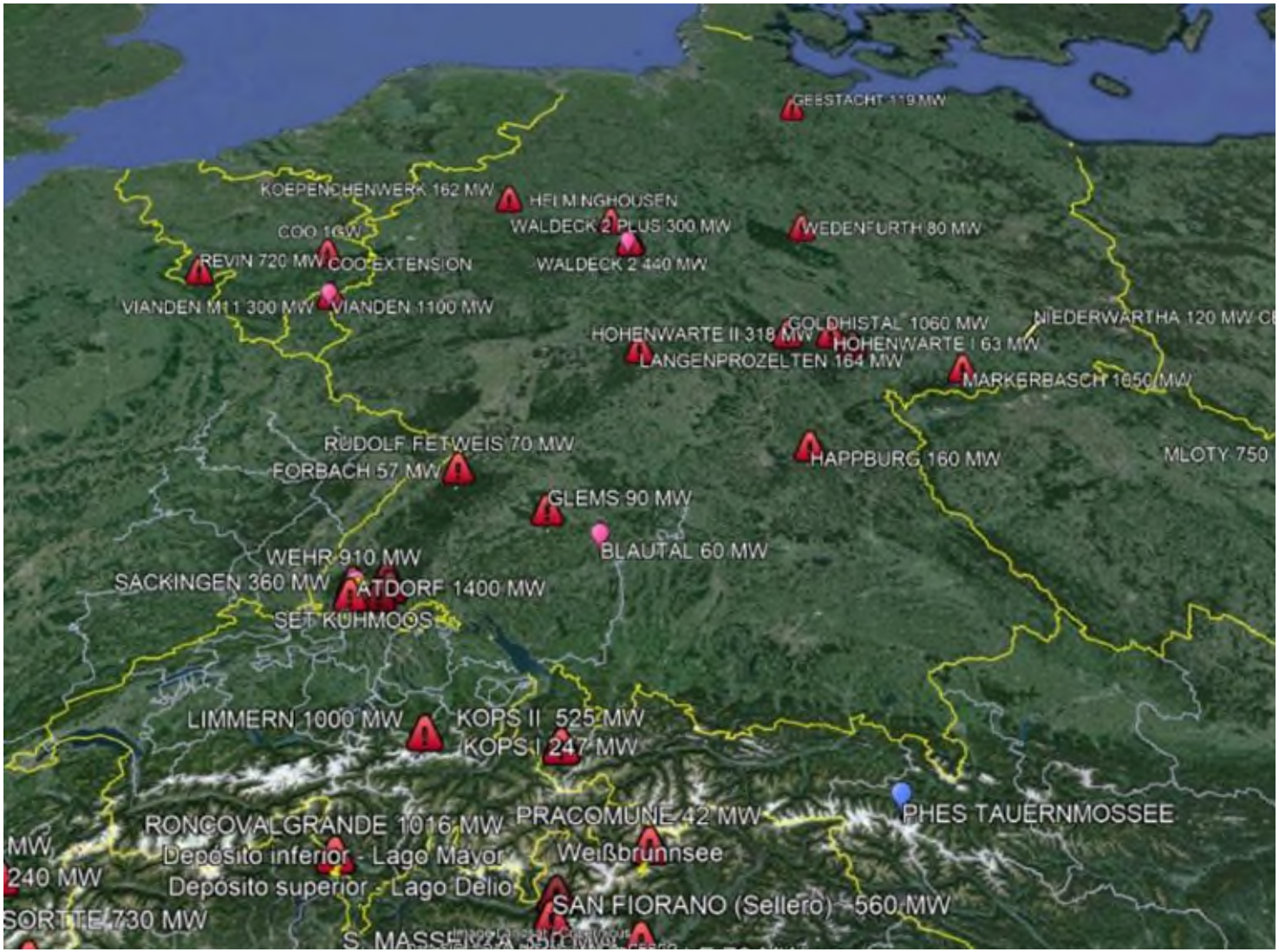
FRANCIA

GRAND MAISON



3.-PROYECTOS AMBITO INTERNACIONAL

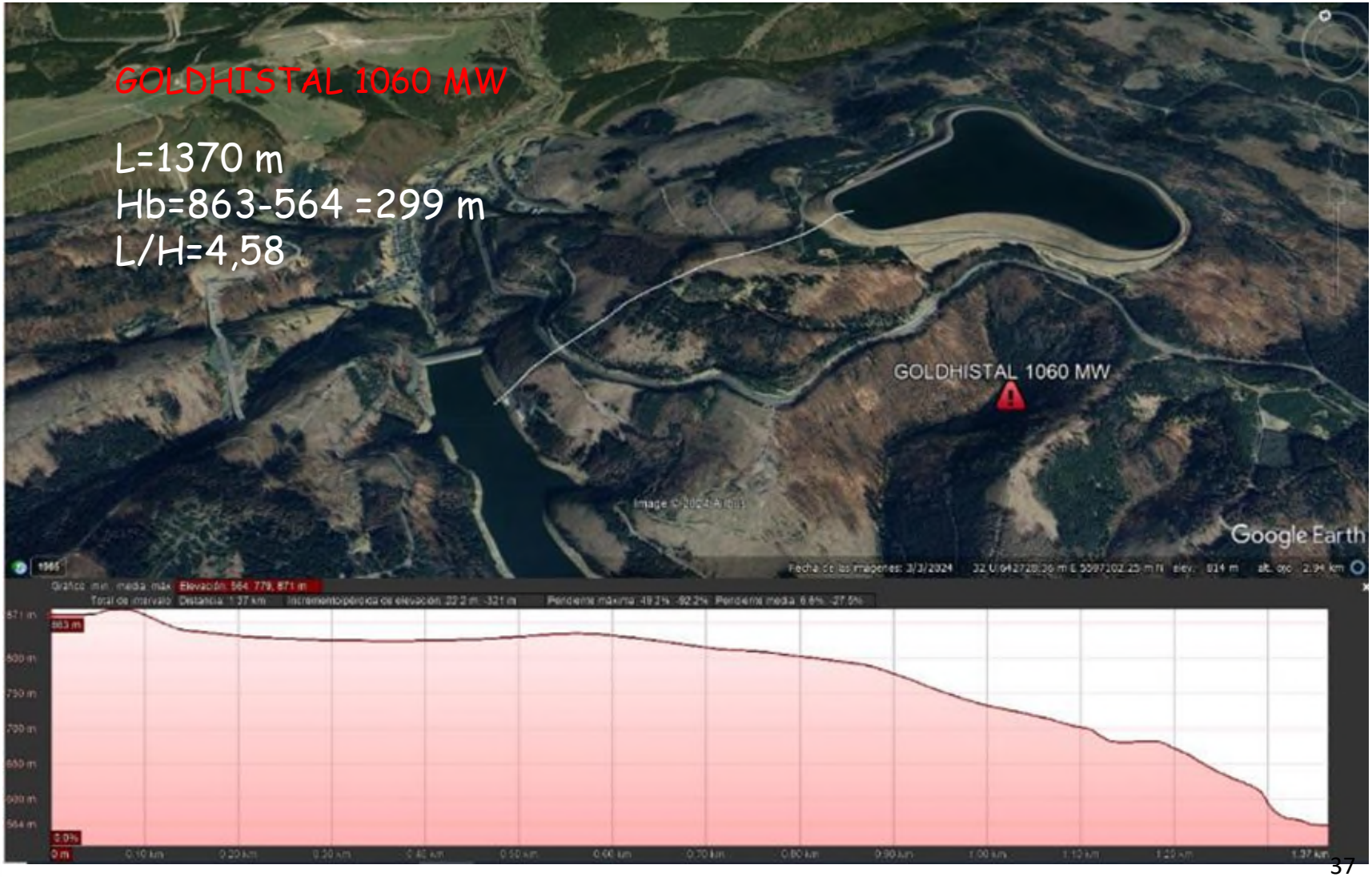
ALEMANIA



3.-PROYECTOS AMBITO INTERNACIONAL

ALEMANIA

GOLDHISTAL



3.-PROYECTOS AMBITO INTERNACIONAL

ALEMANIA

GOLDHISTAL

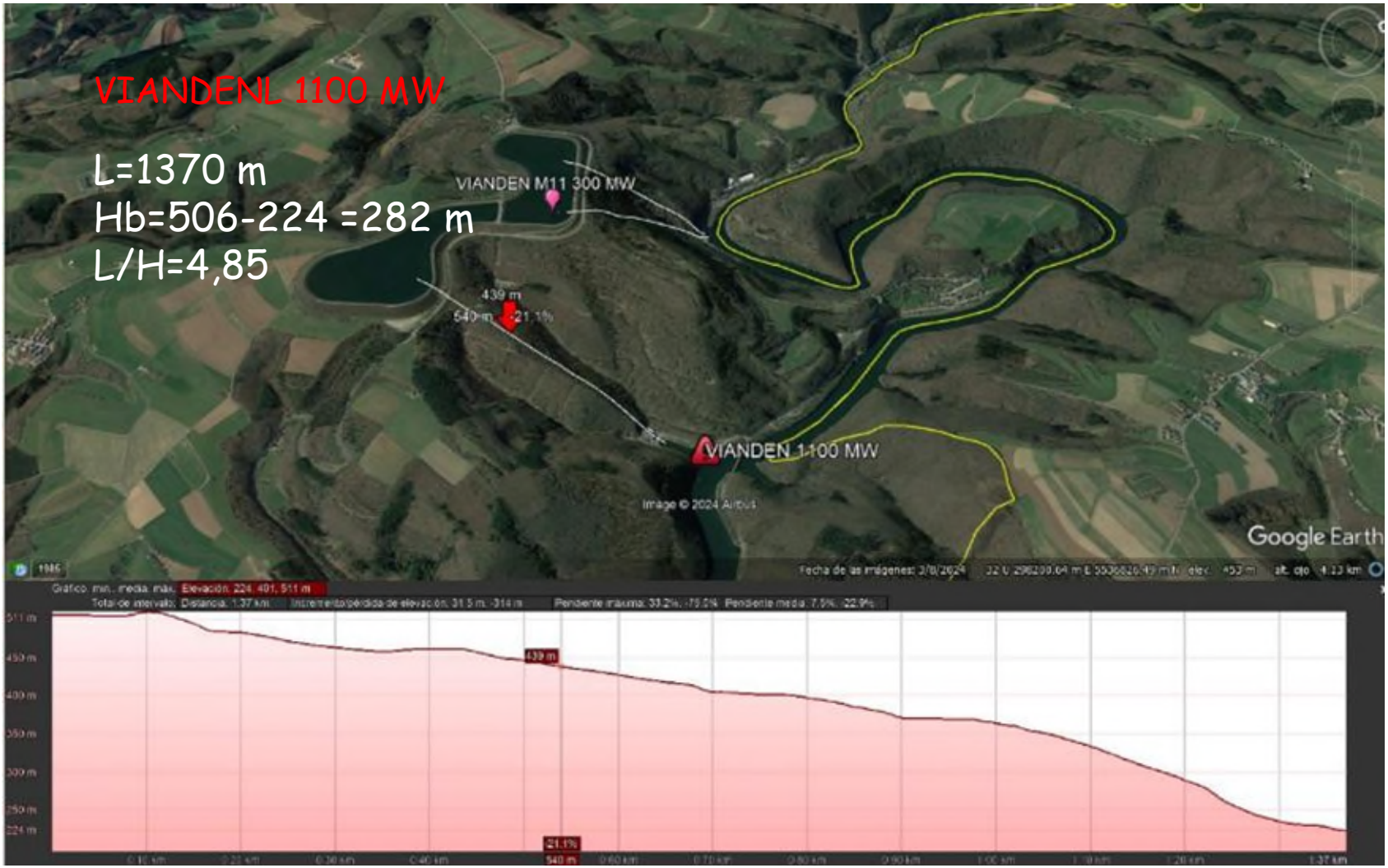




3.-PROYECTOS AMBITO INTERNACIONAL

ALEMANIA-LUXEMBURGO

VIANDEN



3.-PROYECTOS AMBITO INTERNACIONAL

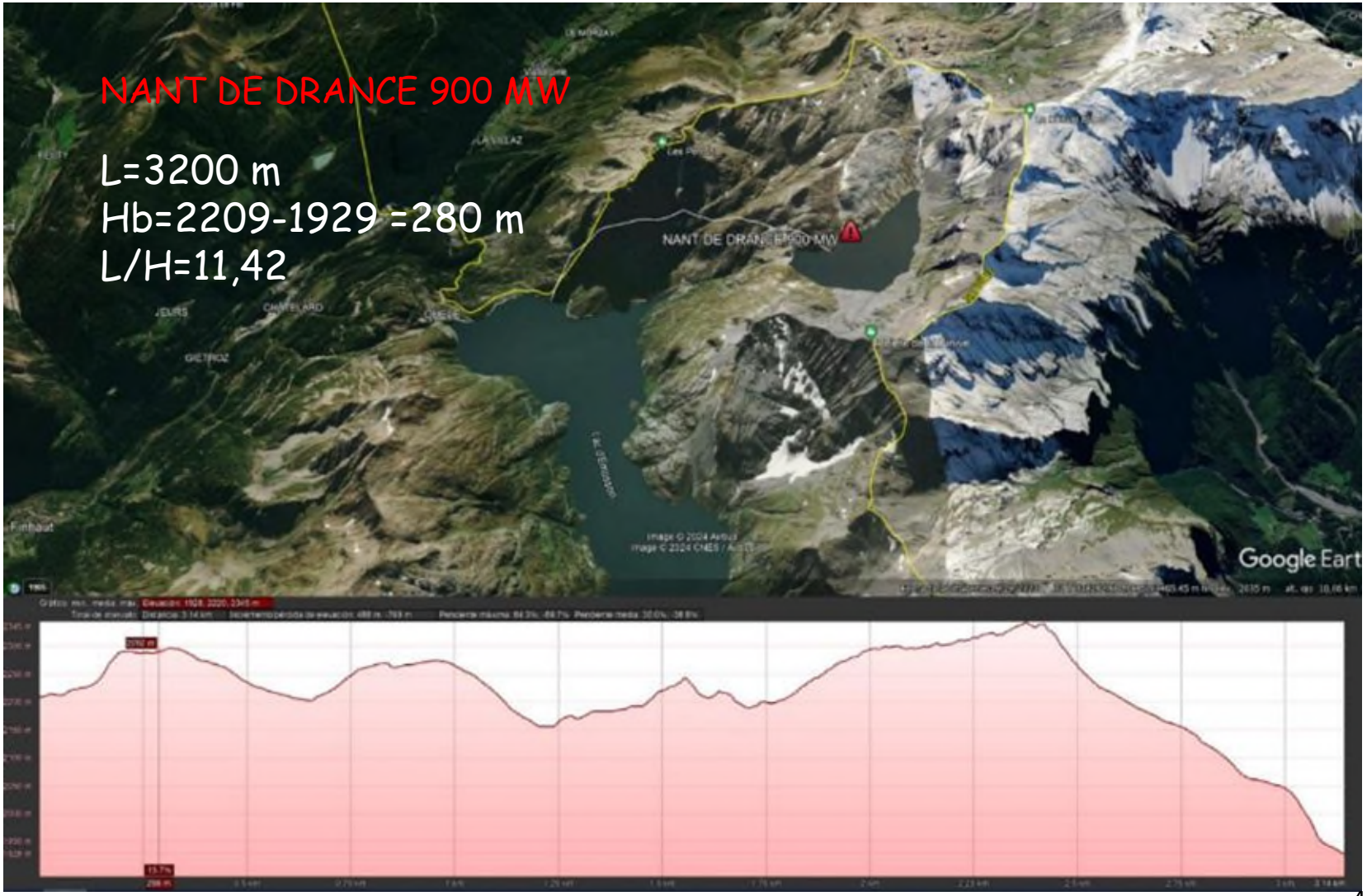
SUIZA



3.-PROYECTOS AMBITO INTERNACIONAL

SUIZA

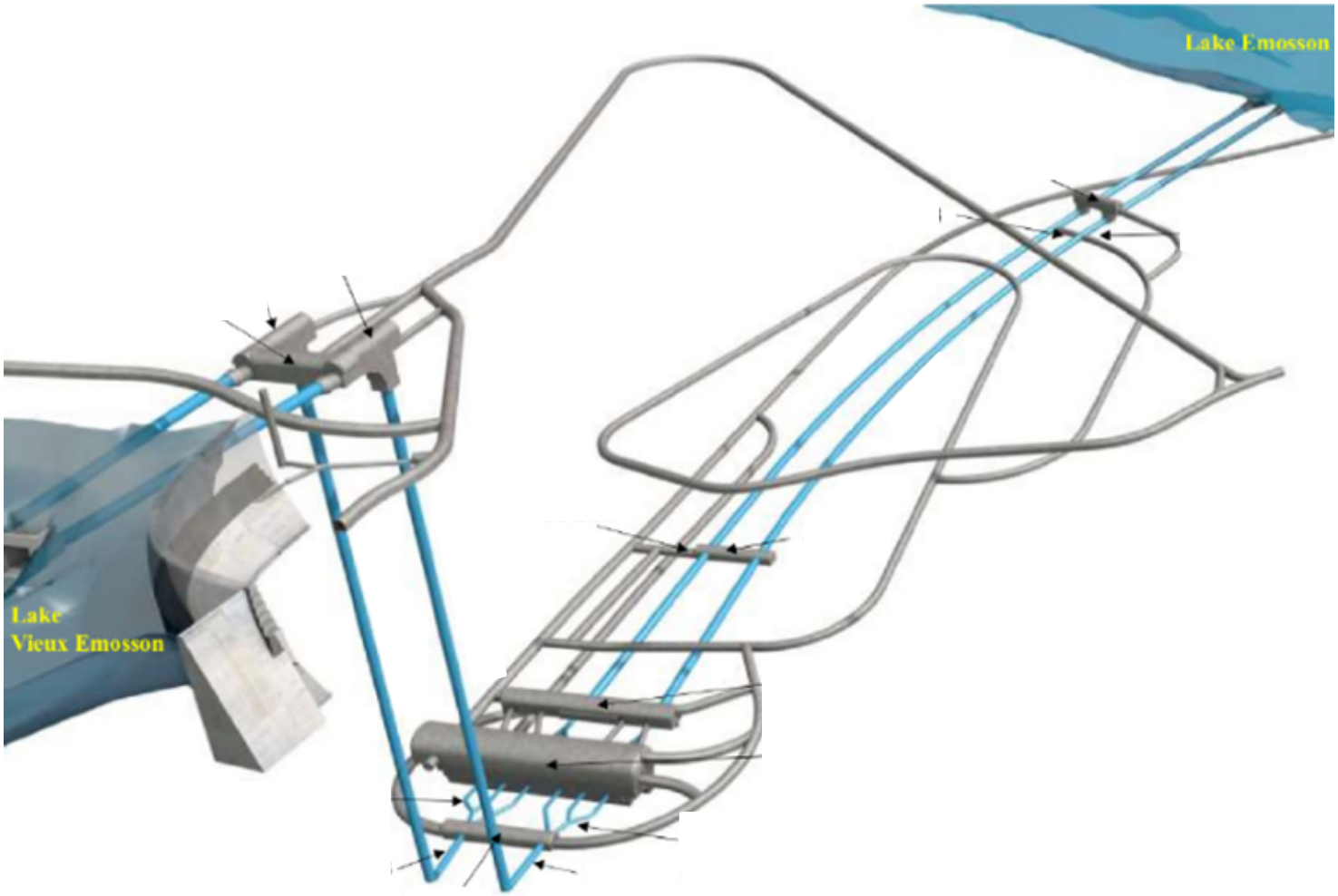
NANT DE DRANCE



3.-PROYECTOS AMBITO INTERNACIONAL

SUIZA

NANT DE DRANCE



3.-PROYECTOS AMBITO INTERNACIONAL

SUIZA

NANT DE DRANCE





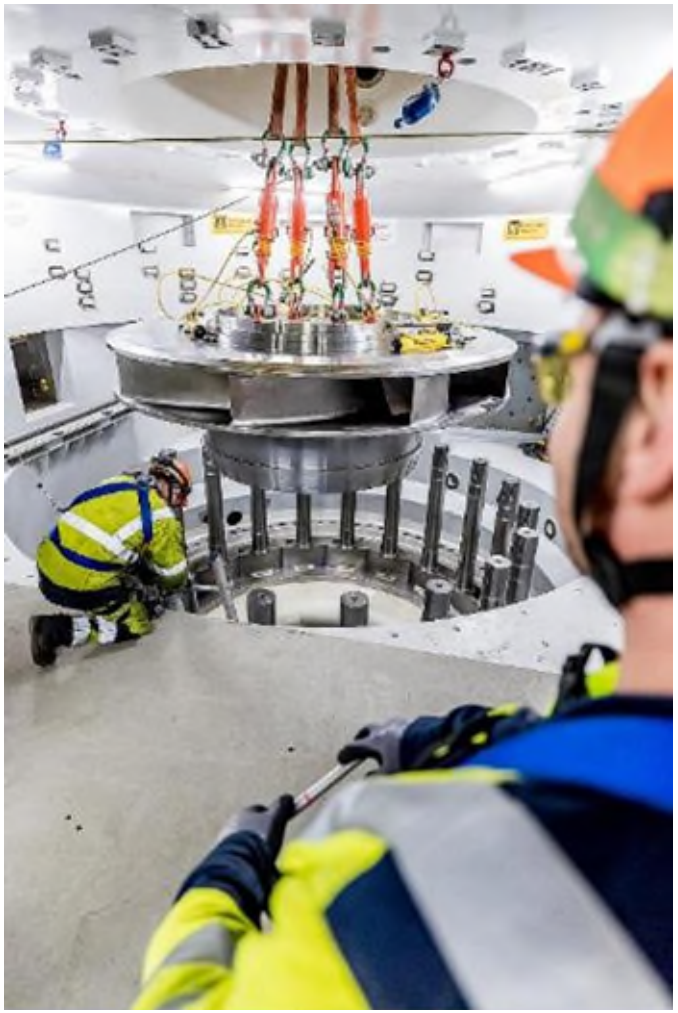
3.-PROYECTOS AMBITO INTERNACIONAL

SUIZA

NANT DE DRANCE







3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRIA



3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRIA

MALTA HAUPSTUFE



3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRIA

MALTA HAUPSTUFE

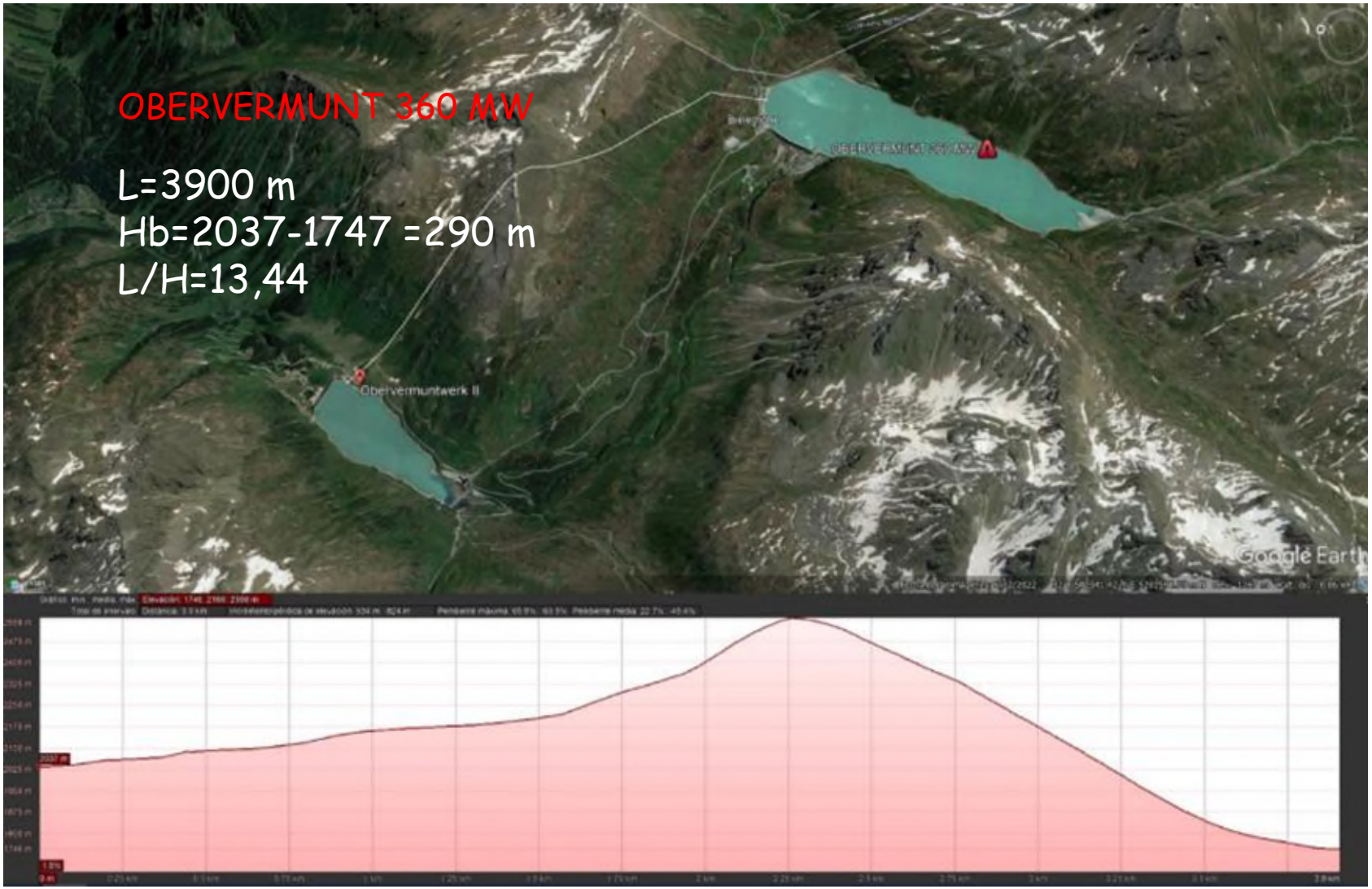


8. 2020 Pranz et al. Pumped Stoage hydroelectricity in Austria

3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRIA

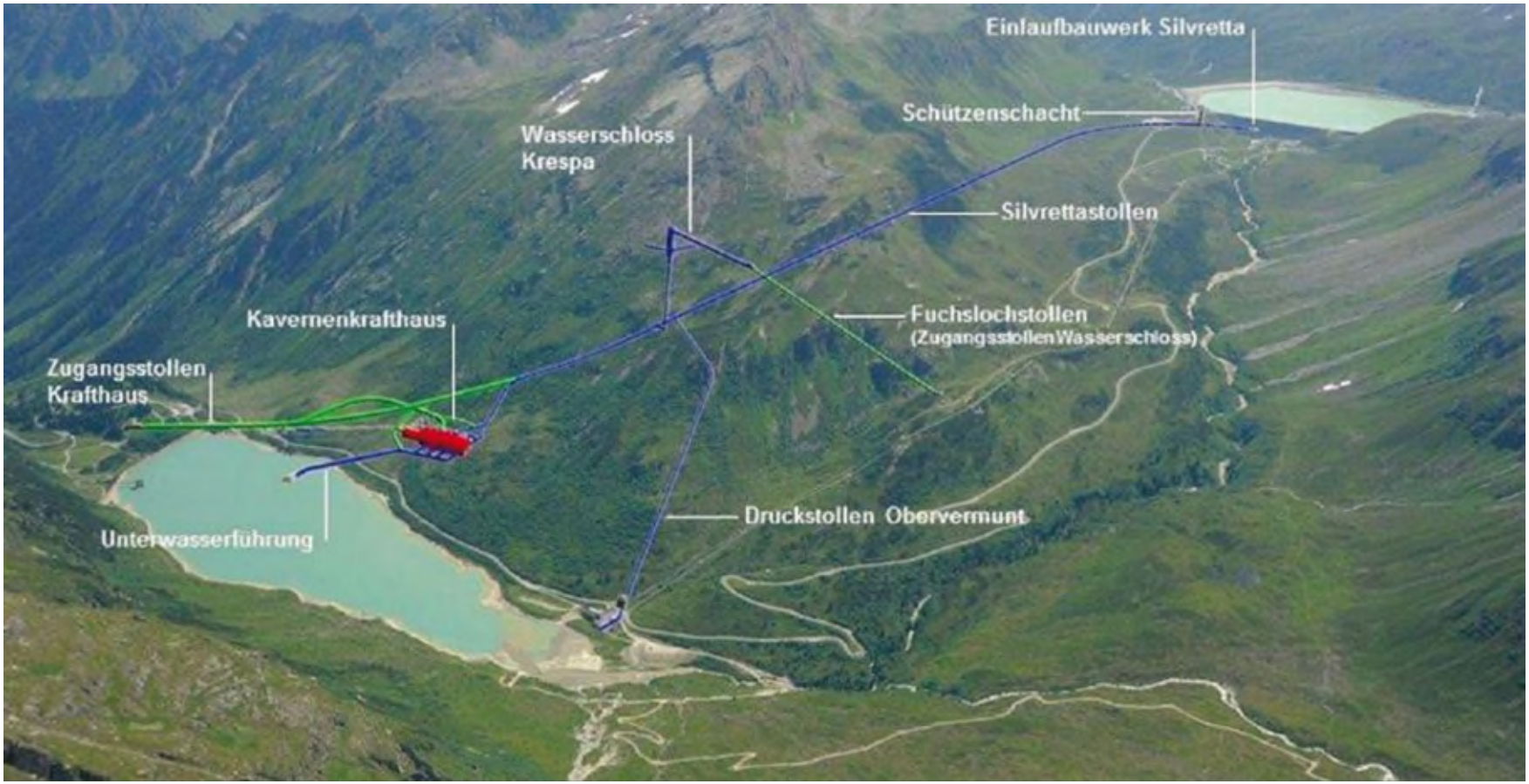
OBERVERMUNT



3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRIA

OBERVERMUNT



3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRIA

OBERVERMUNT



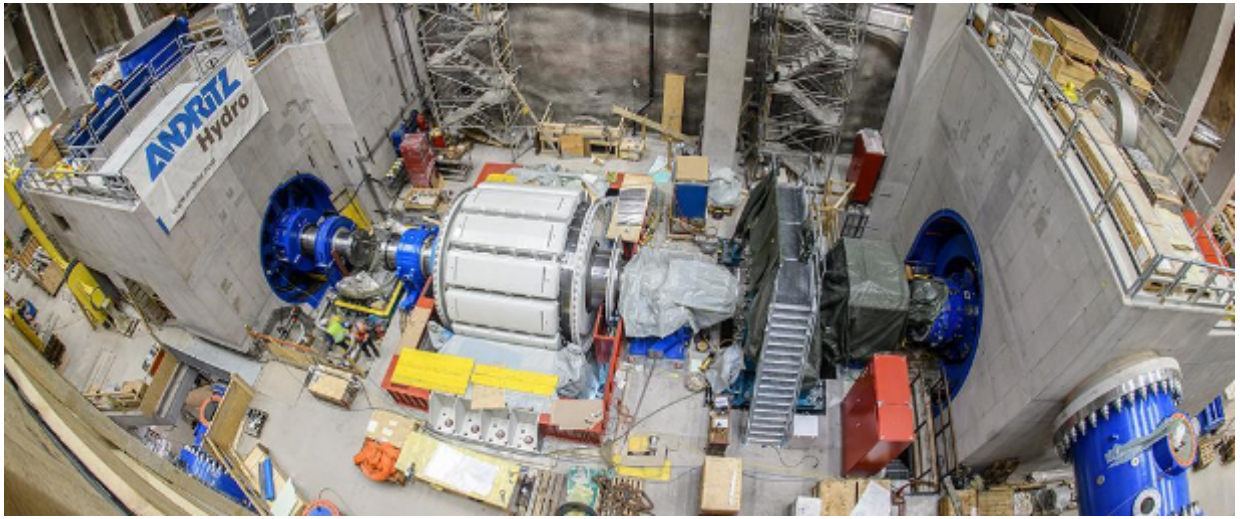
3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRIA

OBERVERMUNT







3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRIA

OBERVERMUNT





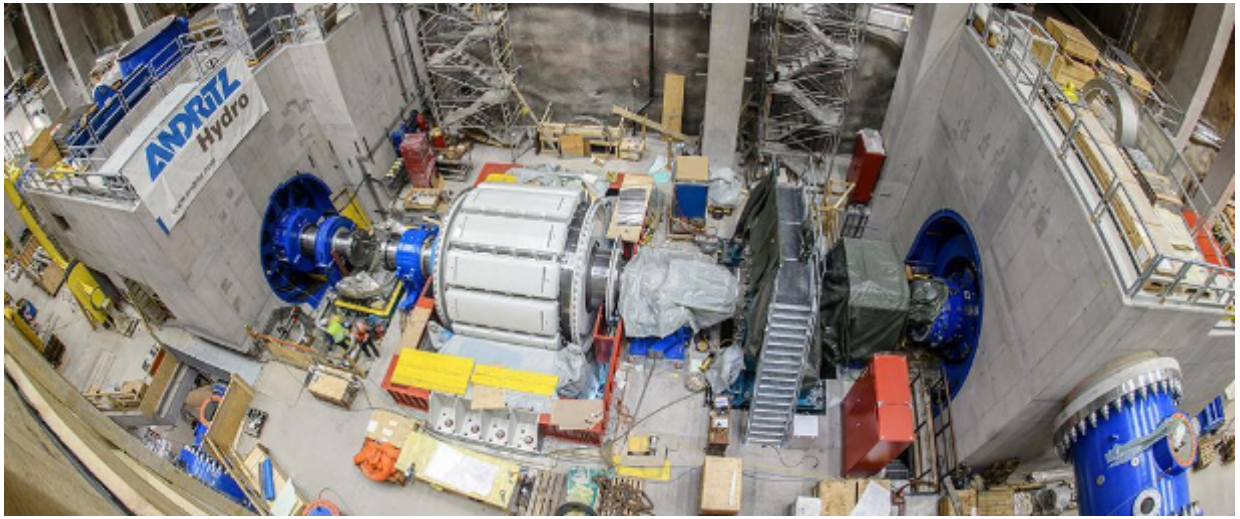
3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRIA

OBERVERMUNT







3.-PROYECTOS AMBITO INTERNACIONAL

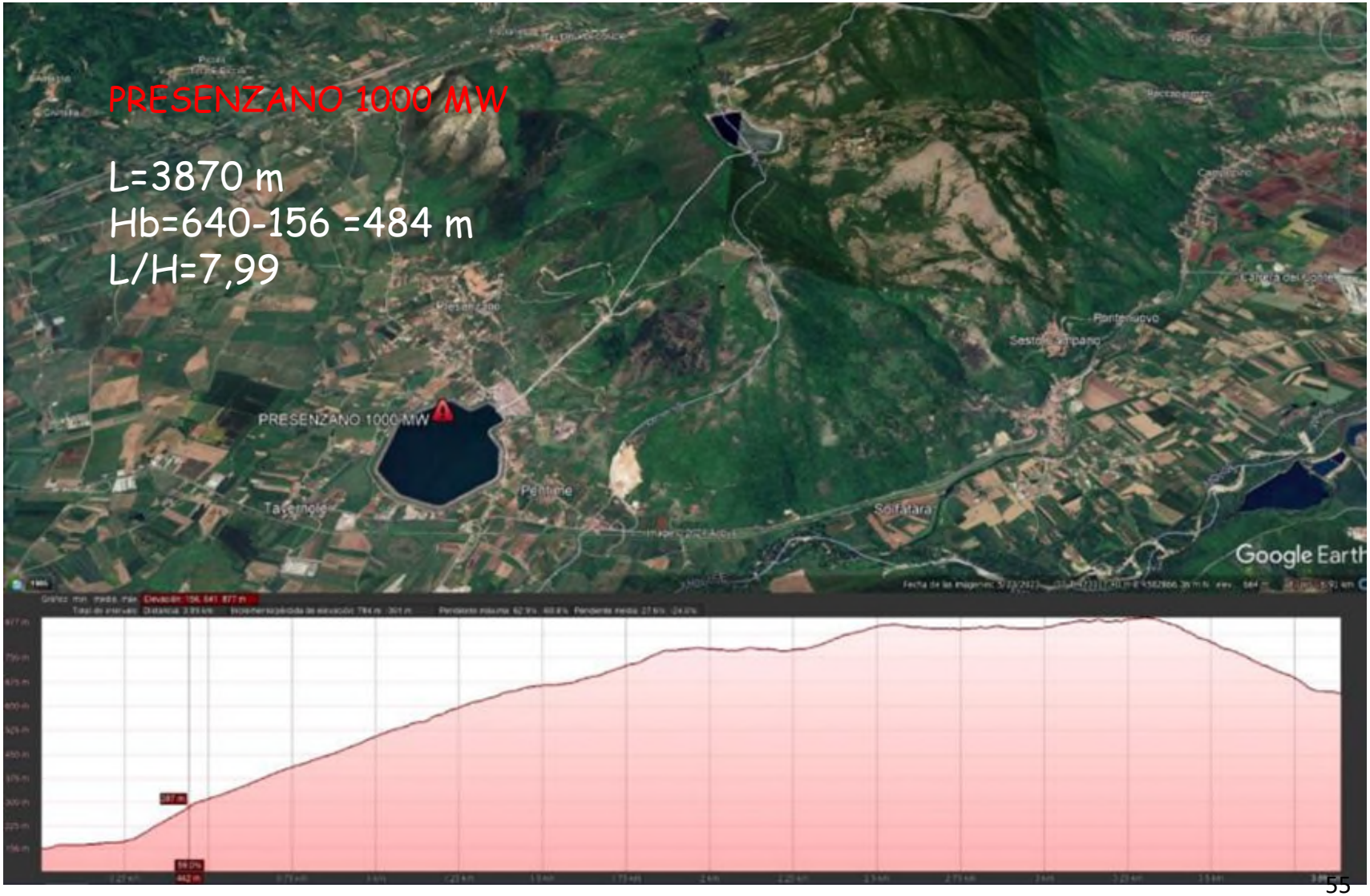
ITALIA



3.-PROYECTOS AMBITO INTERNACIONAL

ITALIA

PRESENZANO



3.-PROYECTOS AMBITO INTERNACIONAL

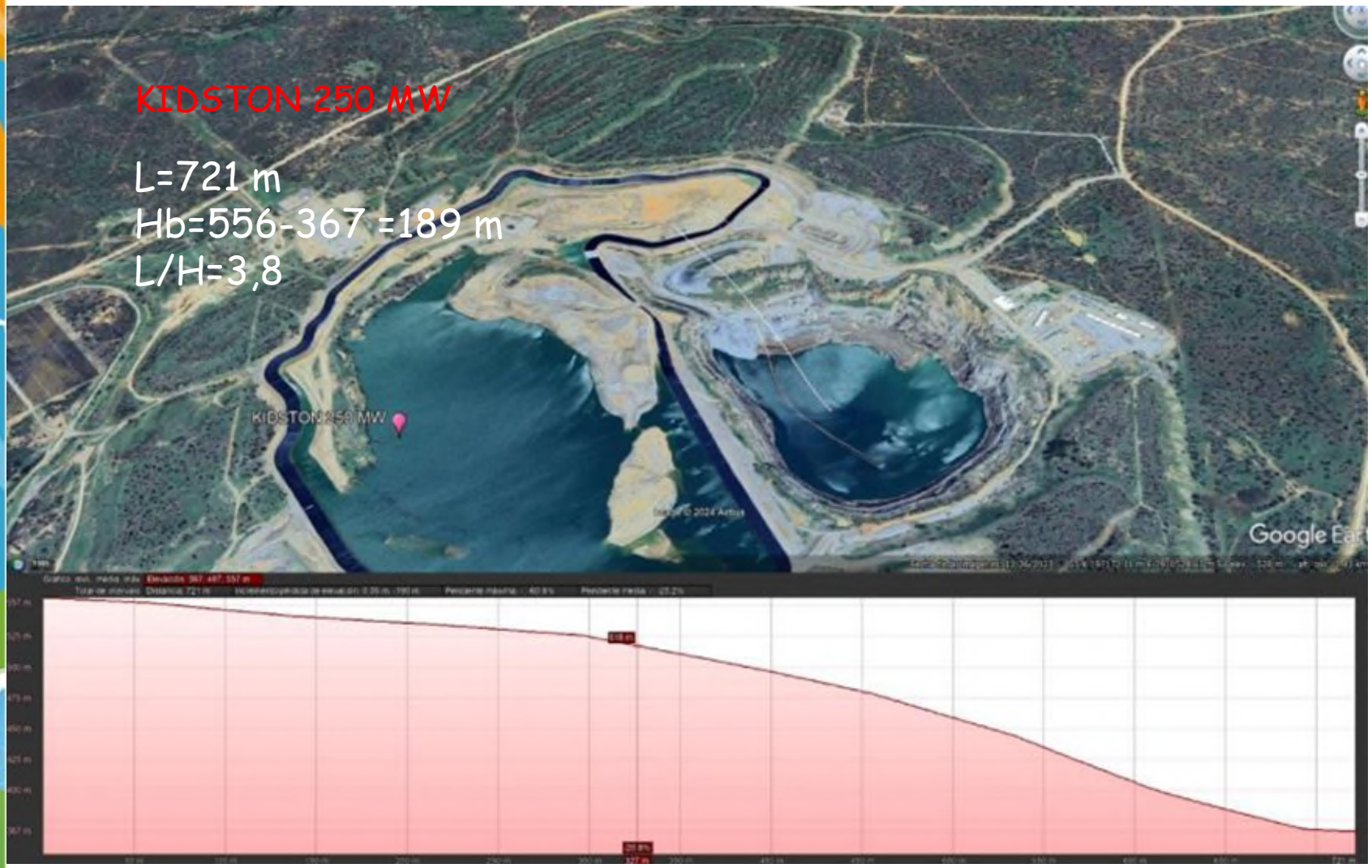
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3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRALIA

KIDSTON



3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRALIA

KIDSTON

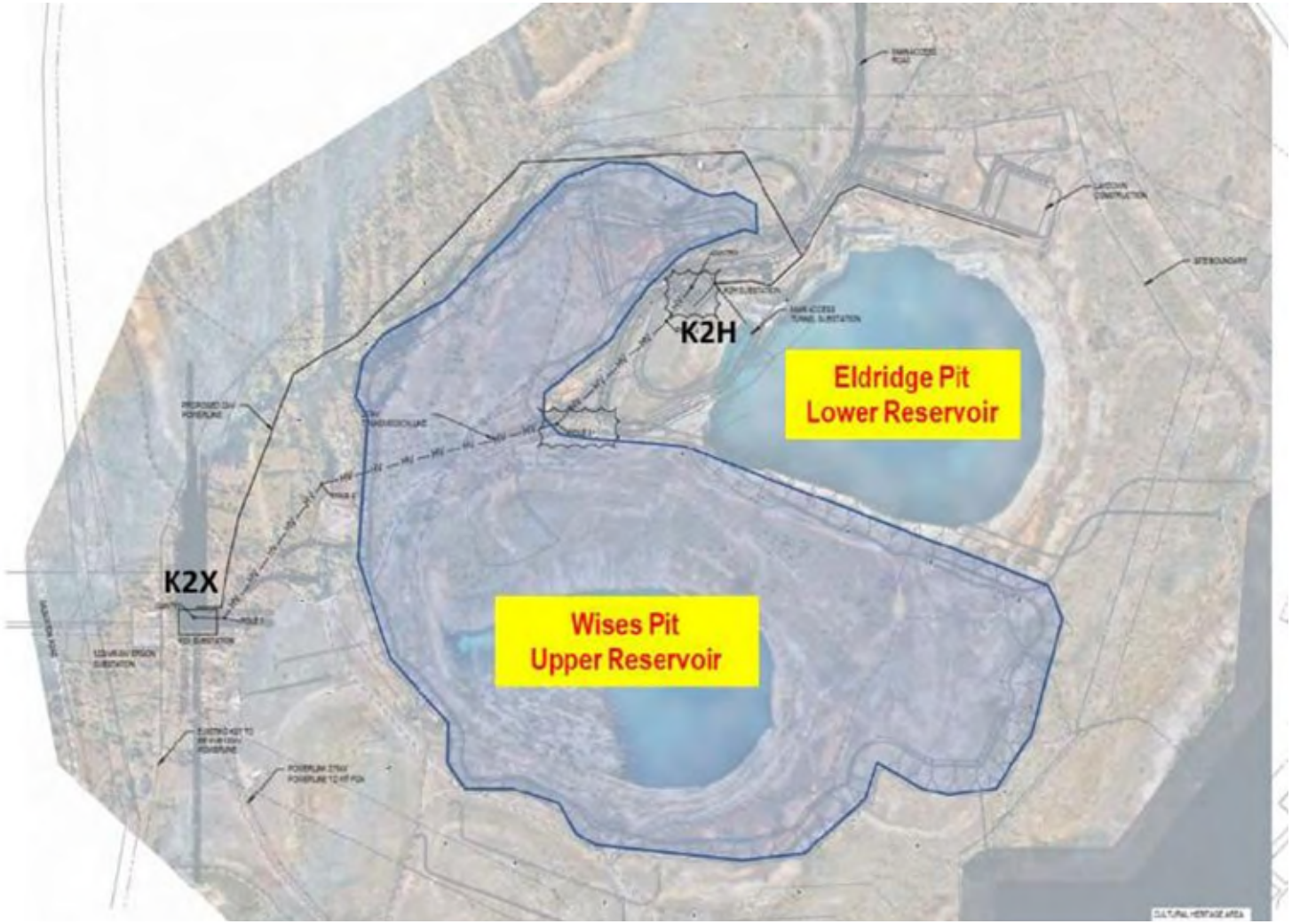


9. 2020 Filtaoff Eureka moment for Kidston Pumped Storage Hydro Project

3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRALIA

KIDSTON

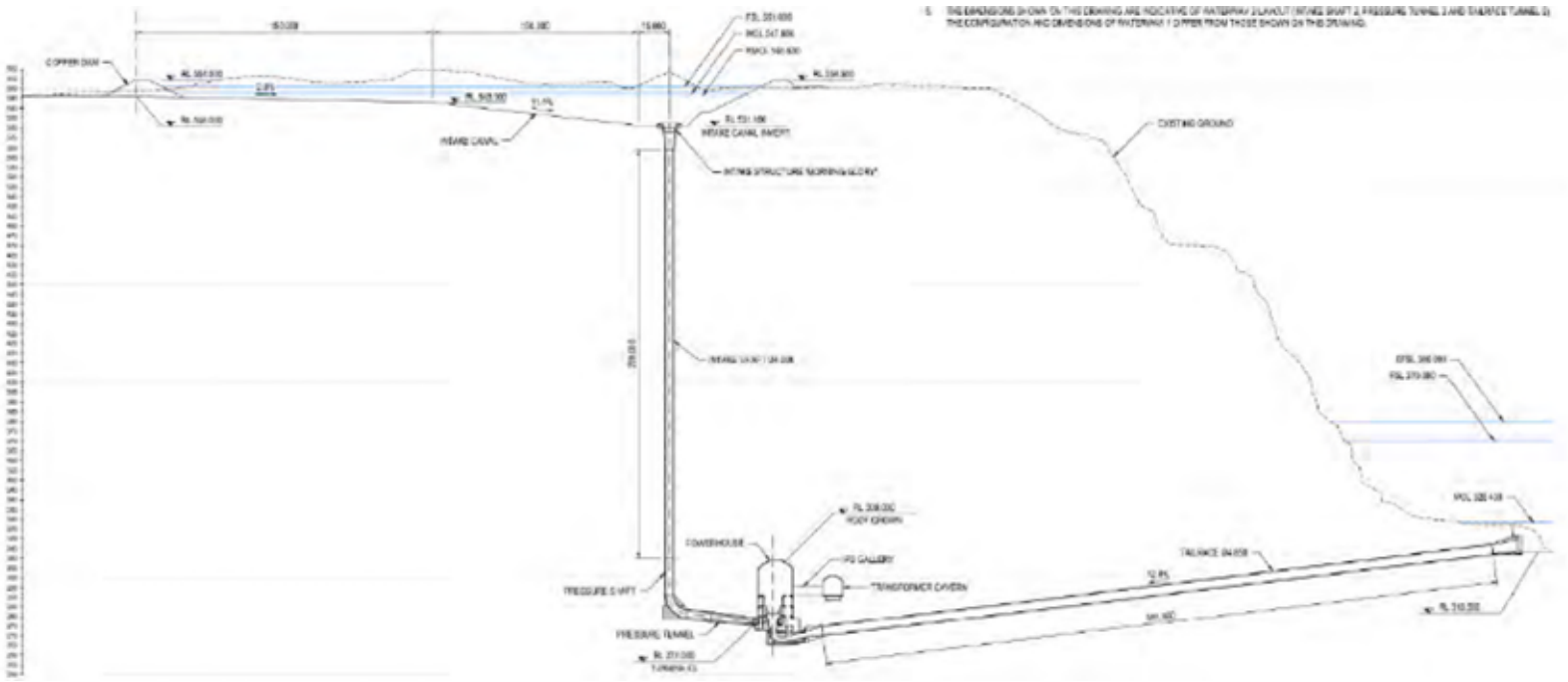


10. 2023 Crowley Kidston Pumped Storage Project- Construction Report

3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRALIA

KIDSTON

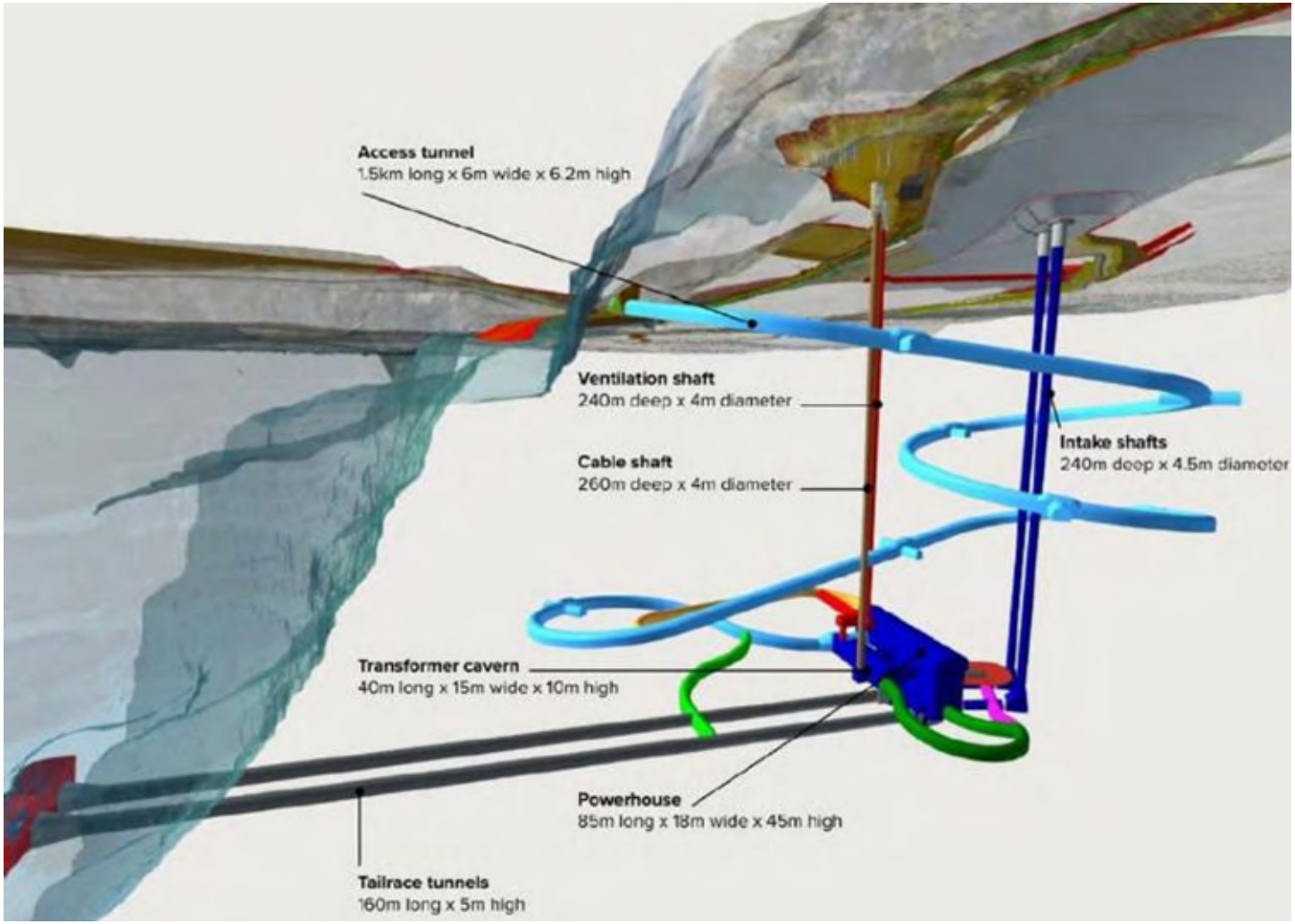


10. 2023 Crowley Kidston Pumped Storage Project- Construction Report

3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRALIA

KIDSTON



10. 2023 Crowley Kidston Pumped Storage Project- Construction Report

3.-PROYECTOS AMBITO INTERNACIONAL

AUSTRALIA

KIDSTON TRANSPORTE

